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Hybrid tree-based machine learning for calorimetric energy calibration using the *Lorenzetti Showers framework*

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ABSTRACT

Accurate and low-latency energy calibration is essential in high-energy physics calorimetry, especially in trigger-oriented environments with stringent timing constraints. Fast calorimetric measurements are affected by pile-up, electronic noise, detector nonlinearities, and shower fluctuations, degrading transverse-energy reconstruction and broadening threshold-based event selection. This work investigates hybrid tree-based machine-learning strategies for event-level calibration within a controlled Lorenzetti simulation benchmark. Datasets were generated with the Lorenzetti Showers framework configured as an ATLAS-inspired electromagnetic calorimeter, and ring-based shower observables were used as compact topological inputs. The proposed approach combines full-ring and dimensionality-reduced representations with Random Forest, Gradient Boosted Decision Tree, residual-learning, and stacked-ensemble regressors to estimate multiplicative calibration factors applied to the standard energy measurement. The best performance was achieved by a hybrid Random Forest model with Gradient Boosted Decision Tree residual correction using the full 100-ring representation. Under the considered benchmark conditions, this configuration reduced RMSE by 60.61%, IQR by 43.39%, relative energy resolution by 44.96%, and the threshold offset required to reach 90% event acceptance by 66.15%. The best compact configuration, a Gradient Boosted Decision Tree using five independent components, achieved reductions of 51.25% in RMSE, 25.45% in IQR, 31.98% in relative energy resolution, and 55.90% in the 90% acceptance threshold offset. The top-ranked configurations also showed low CPU-based trained-model prediction times. Overall, the results support ring-based hybrid tree calibration as a promising benchmark-level strategy for improving simulated fast calorimetric energy estimates, while validation with higher-fidelity simulations, experimental data, and target trigger hardware is required before deployment conclusions.

1. Introduction

Detecting and characterizing rare constituents of matter remains one of the central challenges of particle physics, often requiring high-energy collisions at experimental particle physics accelerator facilities [1]. In this context, high-energy physics (HEP) experiments rely on increasingly sophisticated detector and trigger systems to identify final states of interest amid overwhelming backgrounds. Accurate online and trigger-level energy calibration for electrons and photons is particularly important in analyses involving electroweak and Higgs-boson signatures, where the precision of reconstructed energy directly affects threshold-based event selection, downstream reconstruction quality, and physics sensitivity [2]. Because only a small fraction of recorded interactions contain the targeted signatures, fast and well-calibrated trigger systems are essential for suppressing background-dominated events while preserving those most relevant to the intended analyses.

At the Large Hadron Collider (LHC), the progressive increase in collision energy, luminosity, and event complexity between 2011 and 2026 has further intensified these demands [3,4]. Modern trigger and reconstruction chains must operate under increasingly challenging conditions, including pile-up from multiple simultaneous interactions, electronic noise, and nonlinear detector effects, all of which complicate the accurate estimation of particle energy [5]. These effects are especially relevant for calorimetric measurements, where imperfect calibration may broaden the detector response, introduce bias in

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reconstructed transverse energy, degrade particle identification, and reduce the efficiency and purity of event selection.

From a measurement-science perspective, this problem can be viewed as the construction of a fast data-driven calibration function for a nonlinear, high-throughput measurement system. Similar challenges have motivated the growing use of machine learning in measurement and instrumentation problems, including sensor correction, uncertainty-aware inference, surrogate modeling, and data-driven calibration under operational constraints [6,7]. In such applications, the objective is not only to minimize prediction error, but also to examine the stability of the calibrated response against measurement noise, pile-up contamination, outliers, and detector-response variations across different $(|\eta|, E_T)$ regions. Interpretability, compactness, and computational efficiency are also relevant considerations for future online applications, although practical deployment requires validation beyond the simulation and CPU-based prediction-time studies reported here. These requirements are directly aligned with trigger-level calorimetric reconstruction, where calibration accuracy must be balanced against strict latency and resource constraints.

More broadly, data-driven correction, machine-learning-assisted signal interpretation, and data-driven parameter optimization have become increasingly relevant across measurement and sensing systems. Recent studies have explored machine-learning-assisted interpretation of distributed sensor data for automated monitoring [8], machine-learning-based calibration of low-cost sensing platforms using reference measurements and environmental variables [9], data-driven optimization of sensor configurations and performance [10], and automated self-calibration strategies designed to mitigate sensor drift under resource-constrained operating conditions [11]. Although these systems differ substantially from calorimetric particle detectors in sensing mechanism, physical scale, and application domain, they share a common measurement objective: improving the reliability of inferred quantities in the presence of nonlinear response, noise, interference, uncertainty, drift, and changing operating conditions. The present work addresses this broader challenge in the specific context of simulated fast calorimetric energy calibration under the considered Lorenzetti benchmark.

Efficient event selection and precise energy reconstruction are therefore fundamental requirements in contemporary collider experiments [12,13]. The present work focuses on the calibration of electron transverse energy, a task of direct relevance to final states involving Higgs, Z, and W bosons [14]. Accurate energy estimates are not only important for the correct identification of physical processes, but also for reducing the cost of data storage, transmission, and downstream processing [15,16]. In this scenario, machine-learning (ML) techniques have become increasingly attractive as flexible, data-driven alternatives to conventional calibration procedures.

Among ML approaches, tree-based ensemble methods are particularly appealing for calorimetric calibration because they can model nonlinear response patterns while remaining comparatively efficient at inference time. Previous studies have already shown the potential of Gradient Boosted Decision Trees (BDTs) for energy estimation in HEP applications [17–19]. In the present work, these methods are combined with the Ringer algorithm [14,20], which compresses the information from more than 1000 calorimeter cells into 100 concentric energy rings, providing a compact and trigger-oriented representation of the electromagnetic shower profile.

In a previous study, it has been established the feasibility of Lorenzetti-based machine-learning calibration for fast calorimetric energy reconstruction using ring-encoded shower information [21]. That work evaluated compact feature-selection and feature-extraction strategies, including correlation, mutual information, principal component analysis (PCA), and independent component analysis (ICA), combined with conventional regression models such as multivariate linear regression, polynomial regression, Decision Trees, and BDTs, with and without $(|\eta|, E_T)$ binning. The best configurations demonstrated substantial reductions in root mean squared error (RMSE) and interquartile range (IQR) relative to the Standard FastCalo baseline, with statistically significant improvements in both mean error and variance.

That previous study established reference performance for conventional calibration regressors, including multivariate linear regression, polynomial regression, Decision Trees, and BDTs. The present work extends this baseline by focusing on Random Forest, LightGBM-based Gradient Boosted Decision Trees, residual-correction hybrids, and stacked tree ensembles, evaluated through an expanded measurement-oriented protocol.

The present work extends that study by shifting the focus from conventional regressors and bin-specific training strategies to hybrid tree-based calibration architectures and measurement-oriented performance assessment. In addition to conventional Decision Tree and BDT regressors [22], we investigate Random Forest regression [23], two sequential residual-learning hybrids combining BDT and Random Forest models, and a stacked RF–BDT ensemble with a regularized linear meta-learner [24–26]. Motivated by the use of stacked and hybrid ensembles in nonlinear measurement tasks [27], these architectures are evaluated here as targeted calibration strategies for ring-based calorimetric inputs under constraints relevant to fast HEP reconstruction.

The hybrid architectures are designed to combine complementary tree-based learning mechanisms within the same event-level calibration pipeline. In BDTRF, a LightGBM-based Gradient Boosted Decision Tree first estimates the calibration factor, and a Random Forest subsequently models the residual error left by this prediction. Conversely, RFBDT uses a Random Forest as the first-stage predictor and a LightGBM-based Gradient Boosted Decision Tree for residual correction. The stacked configuration, RFBDTS, trains Random Forest and Gradient Boosted Decision Tree models in parallel and combines their predictions through a Ridge meta-learner.

A second methodological component of this work is the systematic investigation of dimensionality reduction over the 100 ring variables. To this end, we apply both feature-selection and feature-extraction techniques to identify compact representations that preserve most of the predictive information while reducing model complexity. The motivation is twofold: first, to improve calibration efficiency by suppressing redundant or weakly informative inputs; and second, to reduce storage and inference costs, which are especially relevant in latency-constrained environments. The resulting models are evaluated not only in terms of global error metrics, but also with respect to prediction time, number of features, energy resolution, bin-wise stability, and threshold-response behavior.

The dataset used in this study was generated with the Lorenzetti Showers framework [28,29], a general-purpose calorimeter simulation tool configured to reproduce the geometry, materials, and segmentation of the ATLAS electromagnetic calorimeter [30]. This configuration is adopted as a representative and well-documented use case, but it does not correspond to the official ATLAS simulation framework. The use of Lorenzetti offers several practical advantages for ML development: it enables flexible simulation control, provides direct access to low-level calorimeter observables such as cells and ring variables, and facilitates the generation of open datasets with detailed instrumentation-level information that are typically inaccessible outside experimental collaborations. Although the ATLAS electromagnetic calorimeter is used here as the reference scenario, the proposed calibration pipeline could, in principle, be adapted to other calorimetric systems through appropriate detector-specific reconfiguration. However, such applicability is not demonstrated by the present results and would require dedicated validation for each detector.

Compared with previous Lorenzetti-based calibration study [21], the present work differs in four main aspects. First, it introduces hybrid tree-based regression architectures based on residual correction and stacked generalization. Second, it uses $(|\eta|, E_T)$ regions for a posteriori performance evaluation rather than for bin-specific model training.

Third, it expands the assessment beyond RMSE, IQR, and statistical tests by including energy-resolution curves and turn-on behavior around trigger thresholds. Fourth, it introduces a global performance evaluation that jointly accounts for calibration accuracy, regional stability, feature compactness, and CPU-based model prediction time. These extensions reposition the problem as a multi-objective calibration benchmark for simulated fast calorimetric energy estimation.

The main contribution of this study is not the introduction of new standalone machine-learning algorithms. Random Forests, Gradient Boosted Decision Trees, residual learning, and stacked ensembles are established approaches, and BDT-based methods have previously been investigated for energy estimation and calibration in high-energy physics. Rather, this work presents a systematic calibration methodology that evaluates complementary tree-based strategies within a common ring-based calorimetric measurement pipeline. The methodology combines event-wise multiplicative correction, fixed-cardinality feature selection and extraction, globally trained residual-correction and stacked hybrid models, and measurement-oriented performance assessment. The resulting comparison examines the trade-offs between calibration accuracy, error dispersion, energy resolution, trigger-threshold response, regional stability, model compactness, and CPU-based prediction time, thereby identifying benchmark-relevant calibration configurations beyond conventional single-regressor baselines.

The remainder of this paper is organized as follows. Section 2 describes the complete methodology and the simulation setup, including the Lorenzetti Showers configuration, ring-based feature construction, dimensionality-reduction strategies, regression models, and performance-evaluation protocol. Section 3 presents the experimental results, including global accuracy metrics, energy-resolution studies, threshold-response analyses, bin-wise stability, statistical validation, and computational-efficiency comparisons. Section 4 summarizes the main findings and discusses possible directions for future developments.

2. Methodology

The methodology adopted in this work builds on the Lorenzetti-based calibration framework previously introduced in [21]. In that study, the simulation configuration, the ATLAS-inspired electromagnetic calorimeter geometry, the use of single-electron events, the construction of normalized Ringer variables, and the basic event-wise calibration-factor formulation were established and validated for fast calorimetric energy reconstruction. The present work reuses this validated simulation and feature-construction pipeline, while extending the regression stage to hybrid tree-based models and broadening the performance assessment to include energy resolution, turn-on behavior, bin-wise stability, feature compactness, and inference latency.

2.1. Simulation setup and calorimetric observables

The dataset used in this study was generated with the Lorenzetti Showers framework [28,29], using the same ATLAS-inspired electromagnetic calorimeter configuration described in [21]. Lorenzetti provides a self-contained simulation chain that includes event generation, shower propagation in a segmented calorimeter geometry, electronic response emulation, and reconstruction of calorimetric observables. This framework enables direct access to low-level detector information, including calorimeter cells and ring-based energy representations, which are essential for developing and testing machine-learning-based calibration methods.

Approximately 3×10^5 single-electron events from proton–proton collisions were generated using the Lorenzetti built-in event generation pipeline. The simulation assumed a luminosity of $10^{34}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$, consistent with LHC Run 3 conditions [30]. The complete simulated dataset is publicly available at [31], ensuring transparency and reproducibility.

It should be noted that this setup is intended as a controlled and reproducible calorimetric benchmark, not as a replacement for the official ATLAS reconstruction chain. The Lorenzetti configuration preserves key detector-inspired features relevant to this study, including segmentation, electronic noise, pile-up contributions, and access to low-level cell and ring observables. Therefore, the results should be interpreted as a simulation-based assessment of relative calibration performance, with validation on collaboration-grade simulation or experimental data left as a necessary step before deployment.

The Lorenzetti framework was selected because it provides an openly reproducible detector-inspired environment with calorimeter segmentation, electronic noise, pile-up contributions, and access to low-level cell and ring observables. Nevertheless, the present benchmark does not establish detector-independent generalizability or readiness for deployment in an official trigger-reconstruction chain. Validation using collaboration-grade simulation, experimental data, or additional detector-response configurations is required before such conclusions can be drawn.

Electronic noise is included during the pulse-shaping stage of the Lorenzetti simulation, which converts simulated hits into readout signals and emulates the response of calorimeter electronics. The noise contribution follows a Gaussian distribution centered at zero, with layer-dependent standard deviation. Pile-up effects are simulated by merging minimum-bias events into the primary event, following the procedure described in [28].

In this context, pile-up refers to additional detector activity produced by multiple proton–proton interactions occurring within the same, or nearby, readout time window. In the Lorenzetti simulation, this effect is represented by merging minimum-bias events into the primary single-electron event. Minimum-bias events correspond to generic proton–proton collision events generated with loose selection requirements and are commonly used to emulate the diffuse background activity associated with ordinary additional collisions. Nearby calorimetric activity denotes energy deposits from particles close to the electron candidate within, or close to, the region of interest; such activity may originate from pile-up particles or from other particles produced in the same hard interaction.

Fig. 1 shows the distribution of the average pile-up parameter $\langle\mu\rangle$ in the simulated dataset. The distribution contains 10,010 simulated entries, with mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999. Thus, the benchmark used in this work corresponds to a pile-up-inclusive simulation scenario with a significant level of additional collision activity. Accordingly, the performance metrics reported in this work, including RMSE, IQR, energy resolution, and turn-on behavior, were obtained from events containing the pile-up treatment represented in the adopted Lorenzetti configuration. However, the dataset was not generated as a systematic scan over independently controlled pile-up levels. Consequently, the reported results quantify the relative calibration performance of the evaluated models under the simulated pile-up distribution considered here, but do not establish invariance across separately defined low, medium, and high-pile-up regimes.

A cross-sectional view of the simulated calorimeter is shown in Fig. 2, and the corresponding segmentation parameters are summarized in Table 1. The electromagnetic and hadronic calorimeters are segmented into longitudinal layers and preceded by a presampler layer, allowing the simulation to reproduce the main features of a sampling calorimeter relevant to electron energy reconstruction.

Single electrons were generated using a dedicated *electron gun* configuration [32]. This setup injects electrons with predefined energies and incident directions into the detector volume, enabling systematic coverage of a broad kinematic range. This choice is appropriate for calibration studies because it avoids restricting the training sample to specific decay topologies, such as Z-boson decays, whose energy and angular distributions may be narrower. The particle-gun strategy therefore supports the construction of calibration models that remain valid over diverse detector-response conditions [16,32,33].

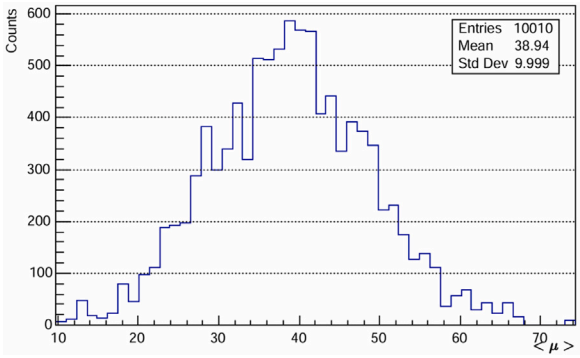


Fig. 1. Distribution of the average pile-up parameter $\langle\mu\rangle$ in the Lorenzetti simulated dataset used in this work. The distribution contains 10,010 simulated entries, with mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999, indicating that the calibration benchmark includes a significant level of additional collision activity.

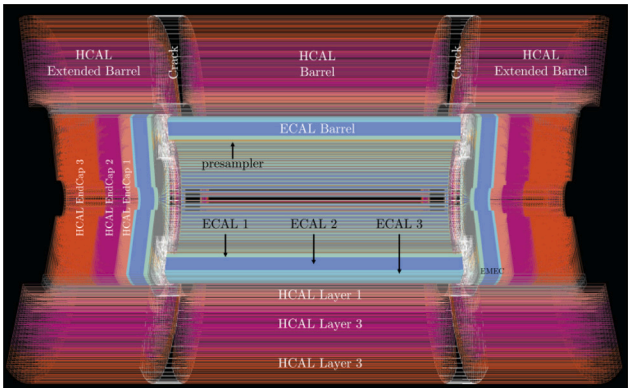


Fig. 2. Cross-sectional view of the general-purpose sampling calorimeter used for simulations with the Lorenzetti Showers framework. The single-electron simulations were constrained to the barrel region for this work.
Source: [28].

For each simulated event, a fixed region of interest (RoI) of 0.4×0.4 in (η,ϕ) is considered. When all seven Lorenzetti calorimeter layers are included, this RoI contains approximately 1000 calorimeter cells per event. To obtain a compact representation of the electromagnetic shower topology, the *Ringer* algorithm [14,20] is applied within the RoI. The algorithm aggregates transverse energy deposits into 100 concentric rings centered on the shower barycenter, as illustrated in Fig. 3. The resulting 100-dimensional vector preserves the radial structure of the shower while reducing the original cell-level dimensionality, making it suitable for low-latency machine-learning calibration [30].

In addition to the normalized ring-energy vector, three physical quantities are used throughout the calibration pipeline:

- E_{true}: the true transverse energy of the generated particle, used as the reference target for calibration;
- E_{fast}: the Standard FastCalo baseline energy estimate, corresponding to the uncalibrated trigger-level energy measurement used for rapid decision-making.
- η: the pseudorapidity of the reconstructed electromagnetic object, used for regional performance evaluation across the detector acceptance.

The global workflow is summarized in Fig. 4. Starting from the Lorenzetti simulation output, the normalized ring variables are used to construct reduced feature representations. These features are then provided to machine-learning regression models that estimate event-level calibration factors. The resulting calibrated energy is finally compared with E_{true} to quantify the improvement over the Standard FastCalo baseline.

Table 1
Segmentation, coverage in $|\eta|$, and cell granularity used in the simulated calorimeter. Electromagnetic and hadronic calorimeters are segmented into three longitudinal layers and preceded by a presampler layer.
Source: [28].

Layer	Sampling	Coverage	Granularity ($\Delta\eta\times\Delta\phi$)
Presampler	Barrel	$0.00 < \lvert\eta\rvert < 1.58$	0.025×0.1
	End-Cap	$1.50 < \lvert\eta\rvert < 1.80$	0.025×0.1
Electromagnetic Calorimeter			
Layer 1	Barrel	$0.00 < \lvert\eta\rvert < 1.55$	0.003×0.1
		$1.37 < \lvert\eta\rvert < 1.80$	0.003×0.1
	End-Cap	$1.80 < \lvert\eta\rvert < 2.00$	0.025×0.1
		$2.00 < \lvert\eta\rvert < 2.37$	0.006×0.1
		$2.37 < \lvert\eta\rvert < 3.20$	0.1×0.1
Layer 2	Barrel	$0.00 < \lvert\eta\rvert < 1.50$	0.025×0.025
	End-Cap	$1.35 < \lvert\eta\rvert < 2.50$	0.025×0.025
		$2.50 < \lvert\eta\rvert < 3.20$	0.1×0.1
Layer 3	Barrel	$0.00 < \lvert\eta\rvert < 1.58$	0.05×0.1
	End-Cap	$1.35 < \lvert\eta\rvert < 2.50$	0.05×0.025
		$2.50 < \lvert\eta\rvert < 3.20$	0.1×0.1
Hadronic Calorimeter			
Layer 1	Barrel	$0.00 < \lvert\eta\rvert < 1.09$	0.1×0.1
	Extended Barrel	$0.94 < \lvert\eta\rvert < 1.77$	0.1×0.1
	End-Cap	$1.50 < \lvert\eta\rvert < 2.50$	0.1×0.1
		$2.50 < \lvert\eta\rvert < 3.20$	0.2×0.2
Layer 2	Barrel	$0.00 < \lvert\eta\rvert < 1.09$	0.1×0.1
	Extended Barrel	$0.85 < \lvert\eta\rvert < 1.41$	0.1×0.1
	End-Cap	$1.50 < \lvert\eta\rvert < 2.50$	0.1×0.1
		$2.50 < \lvert\eta\rvert < 3.20$	0.2×0.2
Layer 3	Barrel	$0.85 < \lvert\eta\rvert < 0.72$	0.2×0.1
	Extended Barrel	$0.85 < \lvert\eta\rvert < 1.41$	0.2×0.1
	End-Cap	$1.50 < \lvert\eta\rvert < 2.50$	0.1×0.1
		$2.50 < \lvert\eta\rvert < 3.20$	0.2×0.2

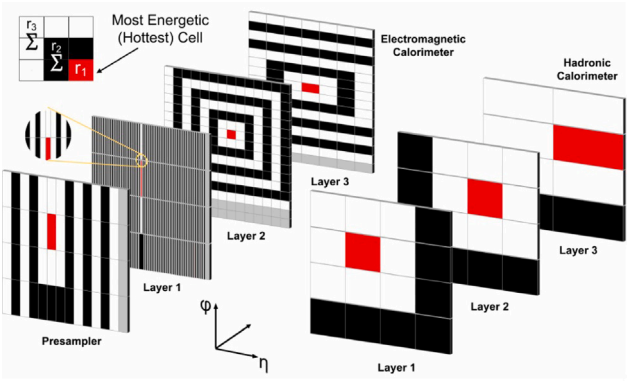


Fig. 3. Graphical representation of the ring-based reconstruction concept. The Ringer algorithm centers the reconstruction on the highest-energy cell and groups surrounding calorimeter cells into concentric rings across different calorimeter layers. Each ring aggregates the transverse energy deposited by the cells it encompasses, preserving the spatial topology of the electromagnetic shower in a one-dimensional profile.
Source: [34,35].

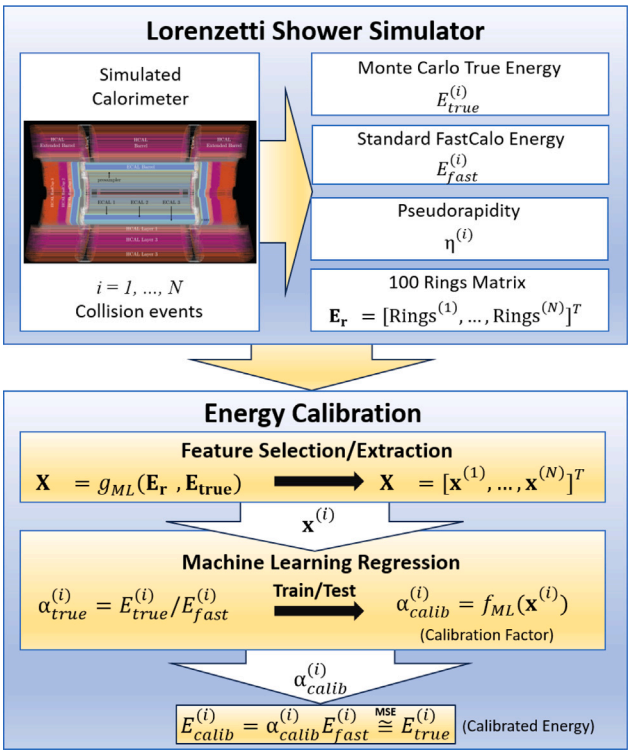


Fig. 4. Schematic workflow of the proposed energy-calibration system integrated with Lorenzetti Showers. The feature-selection/extraction and machine-learning regression blocks constitute the main processing stages used to estimate the event-level calibration factors.

2.2. Standard FastCalo energy and ring encoding

For completeness, this section summarizes the main definitions used to construct the Standard FastCalo energy estimate and the ring-based calorimetric representation. A detailed description of the same Lorenzetti-based reconstruction pipeline was previously reported in [21]. In the present work, these definitions are retained to ensure a self-contained description of the calibration procedure, while the methodological extension is introduced in the subsequent regression and performance-evaluation stages.

Starting from the raw calorimeter signals, the energy is reconstructed at the cell level, aggregated into clusters, corrected for detector effects and pile-up, and finally transformed into structured ring features suitable for machine-learning-based energy calibration [30]. The reconstructed energy in the kth calorimeter cell is estimated using optimal filtering of the digitized signal:

$$E_k^{\text{cell}} = \sum_{j=1}^{N_s} w_j S_{kj}, \tag{1}$$

where S_{kj} is the signal sampled at time j, w_j are the optimal-filter weights, and N_s is the number of signal samples [30].

The reconstructed cell energies are aggregated within an adaptively defined electromagnetic shower cluster to obtain the raw cluster energy:

$$E_{\text{raw}} = \sum_{k\in\text{cluster}} E_k^{\text{cell}}. \tag{2}$$

The transverse component of the fast energy estimate, hereafter referred to as Standard FastCalo energy, is then computed as

$$E_{\text{fast}} = \frac{E_{\text{raw}}}{\cosh(\eta)}, \tag{3}$$

where η is the pseudorapidity of the cluster centroid.

To construct a compact topological representation of the electromagnetic shower, the *Ringer* algorithm aggregates the transverse energy deposits into concentric rings centered on the shower core, as illustrated in Fig. 3. The transverse energy associated with the jth ring is defined as

$$E_{r,j} = \sum_{k\in\text{ring}\,j} \frac{E_k^{\text{cell}}}{\cosh(\eta)}, \tag{4}$$

where E_k^{cell} is the reconstructed energy in the kth calorimeter cell belonging to ring j.

The ring energies are normalized to form a dimensionless shower-shape profile:

$$\tilde{E}_{r,j} = \frac{E_{r,j}}{\sum_{\ell=1}^{100} E_{r,\ell}}, \tag{5}$$

where $\tilde{E}_{r,j}$ is the normalized transverse energy of the jth ring. This normalization emphasizes the spatial distribution of the shower rather than its absolute magnitude, allowing the calibration model to exploit topological information independently of the original energy scale.

Although normalization reduces the dependence of the ring representation on the absolute shower-energy scale, it does not make the profile invariant to spatial topological disturbances. Nearby calorimetric activity, including pile-up-induced deposits or partially overlapping showers from nearby particles, can redistribute energy across the rings and modify the normalized shower shape, particularly in outer-ring regions. Therefore, the normalized ring representation should be interpreted as a compact shower-topology descriptor, not as an intrinsically pile-up-invariant observable.

For a dataset with N events, the normalized ring-energy matrix is written as

$$\tilde{\mathbf{E}}_r = \begin{bmatrix} \tilde{E}^{(1)}_{r,1} & \cdots & \tilde{E}^{(1)}_{r,100} \\ \tilde{E}^{(2)}_{r,1} & \cdots & \tilde{E}^{(2)}_{r,100} \\ \vdots & \ddots & \vdots \\ \tilde{E}^{(N)}_{r,1} & \cdots & \tilde{E}^{(N)}_{r,100} \end{bmatrix}, \tag{6}$$

where each row corresponds to one event and encodes the normalized shower profile across the 100 rings. For notational convenience, the full ring vector of the ith event is denoted by

$$\text{Rings}[1:100]^{(i)} = \left[\tilde{E}^{(i)}_{r,1}, \ldots, \tilde{E}^{(i)}_{r,100}\right]. \tag{7}$$

The Standard FastCalo estimate E_{fast} provides the initial trigger-level transverse-energy measurement. Because this estimate is affected by detector response, noise, pile-up, and resolution effects, a data-driven calibration factor is introduced to obtain a corrected energy estimate closer to E_{true}.

2.3. Proposed calibration method

The calibration procedure follows the event-wise multiplicative correction strategy introduced in [21]. For the ith event, the calibrated transverse energy is defined as

$$E^{(i)}_{\text{calib}} = \alpha^{(i)}_{\text{calib}} E^{(i)}_{\text{fast}}, \tag{8}$$

where $\alpha^{(i)}_{\text{calib}}$ is the calibration factor estimated by a machine-learning model.

The target calibration factor is obtained from the true and fast transverse-energy estimates:

$$\alpha^{(i)}_{\text{true}} = \frac{E^{(i)}_{\text{true}}}{E^{(i)}_{\text{fast}}}. \tag{9}$$

Thus, the calibration problem is formulated as a supervised regression task in which the model estimates $\alpha^{(i)}_{\text{calib}}$ from a reduced representation of the normalized ring-energy vector.

Let

$$\mathbf{X} = g_{\mathrm{ML}}(\tilde{\mathbf{E}}_r, \mathbf{E}_{\mathrm{true}}) \in \mathbb{R}^{N\times F} \tag{10}$$

be the reduced feature matrix obtained from the normalized ring-energy matrix $\tilde{\mathbf{E}}_r$, where F is the number of selected or extracted features. The ith event is represented by

$$\mathbf{x}^{(i)} = \left[x_1^{(i)}, \ldots, x_F^{(i)}\right] \in \mathbb{R}^F. \tag{11}$$

A regression model $f_{\mathrm{ML}}(\cdot)$ is then trained to estimate the calibration factor:

$$\alpha_{\mathrm{calib}}^{(i)} = f_{\mathrm{ML}}(\mathbf{x}^{(i)}). \tag{12}$$

The final calibrated energy is therefore given by

$$E_{\mathrm{calib}}^{(i)} = f_{\mathrm{ML}}(\mathbf{x}^{(i)}) E_{\mathrm{fast}}^{(i)} \approx E_{\mathrm{true}}^{(i)}. \tag{13}$$

During training, the model parameters are optimized to minimize the mean squared error between the calibrated and true transverse energies:

$$\mathrm{MSE}(E) = \frac{1}{N}\sum_{i=1}^{N}\left(E_{\mathrm{true}}^{(i)} - f_{\mathrm{ML}}(\mathbf{x}^{(i)}) E_{\mathrm{fast}}^{(i)}\right)^2. \tag{14}$$

This formulation defines the calibration pipeline used throughout the paper: $g_{\mathrm{ML}}(\cdot)$ constructs a compact representation from the normalized ring energies, and $f_{\mathrm{ML}}(\cdot)$ estimates the event-level correction factor applied to the Standard FastCalo energy. The main methodological extensions of the present work lie in the fixed-cardinality feature-reduction strategy, the hybrid tree-based regressors, and the expanded performance-evaluation protocol described in the following subsections.

2.4. Feature selection and extraction

Feature selection and extraction are used to reduce the original 100-dimensional ring representation while preserving the information most relevant to the calibration target. In the previous Lorenzetti-based calibration study [21], correlation- and mutual-information-based strategies were defined using fixed relevance thresholds, which led to feature subsets with different cardinalities depending on the threshold value. In contrast, the present work adopts a fixed-cardinality ranking strategy: for each feature-selection or feature-extraction method, the candidate variables are ordered according to their relevance to E_{true}, and the top

$$m \in \{50, 10, 5, 2\}$$

features or components are retained. This change enables a controlled comparison between CORR, MUTI, PCA, and ICA under the same input dimensionalities, which is important for assessing the trade-off between calibration accuracy, model compactness, and inference latency.

Let $\tilde{\mathbf{E}}_r \in \mathbb{R}^{N\times 100}$ denote the normalized ring-energy matrix. The dimensionality-reduction operator

$$\mathbf{X} = g_{\mathrm{ML}}(\tilde{\mathbf{E}}_r, \mathbf{E}_{\mathrm{true}}) \in \mathbb{R}^{N\times F} \tag{15}$$

maps the original ring representation into a reduced feature matrix $\mathbf{X}$, where $F = m$ for all reduced configurations considered in this study. Four strategies are evaluated: two feature-selection methods, based on Pearson correlation and mutual information, and two feature-extraction methods, based on principal component analysis and independent component analysis [22,36,37].

For correlation-based selection, each original ring variable is scored according to its Pearson correlation coefficient with the true transverse energy:

$$\rho_j = \rho(\tilde{\mathbf{E}}_{r,j}, \mathbf{E}_{\mathrm{true}}), \qquad j = 1, \ldots, 100, \tag{16}$$

where $\tilde{\mathbf{E}}_{r,j}$ denotes the jth column of $\tilde{\mathbf{E}}_r$. The ring variables are ranked by $|\rho_j|$, and the top m variables are retained. The resulting configurations are denoted by CORR-m.

For mutual-information-based selection, each ring variable is scored using its mutual information with the true transverse energy:

$$I_j = \mathrm{MI}(\tilde{\mathbf{E}}_{r,j}, \mathbf{E}_{\mathrm{true}}), \qquad j = 1, \ldots, 100. \tag{17}$$

The variables are ranked by I_j, and the top m variables are retained. These configurations are denoted by MUTI-m. Unlike the correlation score, mutual information can capture nonlinear dependencies between the ring response and the calibration target [38].

For principal component analysis, the normalized ring matrix is transformed into an orthogonal basis of uncorrelated components. The resulting components are ordered by their relevance to the calibration target, measured through the absolute Pearson correlation with $\mathbf{E}_{\mathrm{true}}$:

$$|\rho(\mathbf{PCA}^{(1)}, \mathbf{E}_{\mathrm{true}})| \geq \cdots \geq |\rho(\mathbf{PCA}^{(m)}, \mathbf{E}_{\mathrm{true}})|. \tag{18}$$

The top m components define the PCA-m configurations.

For independent component analysis, the normalized ring matrix is decomposed into statistically independent latent components. As with PCA, the components are ranked according to their absolute correlation with the true transverse energy:

$$|\rho(\mathbf{ICA}^{(1)}, \mathbf{E}_{\mathrm{true}})| \geq \cdots \geq |\rho(\mathbf{ICA}^{(m)}, \mathbf{E}_{\mathrm{true}})|. \tag{19}$$

The top m components are retained to form the ICA-m configurations. This representation is useful for investigating whether non-Gaussian latent structures in the shower profile provide compact information for calibration [37].

The notation used throughout the paper is therefore as follows: CORR-m and MUTI-m denote the top m original ring variables selected by correlation and mutual information, respectively, whereas PCA-m and ICA-m denote the top m extracted components retained after target-relevance ordering. This fixed-cardinality design makes it possible to compare feature-selection and feature-extraction methods under identical dimensionality constraints.

To prevent information leakage, all data-dependent feature-selection and feature-extraction operations were fitted exclusively on the corresponding training data. In particular, CORR and MUTI rankings were computed using only the training partition, whereas PCA and ICA were fitted on the training ring-energy matrix and their components were ranked using only the associated training values of E_{true}. The selected variables, fitted transformations, and component orderings were subsequently applied unchanged to the corresponding validation or held-out test data.

For PCA and ICA, the variance associated with each retained component is also analyzed to support interpretability. In PCA, the explained variance is obtained from the eigenvalues of the covariance or correlation matrix. In ICA, although the components are not inherently ordered by variance, a post-hoc variance analysis is used to assess the relative contribution of the retained components. The final reduced matrix $\mathbf{X} \in \mathbb{R}^{N\times F}$ is then used as input to the regression models described in the next section.

2.5. Machine learning for regression

The calibration task is formulated as a supervised regression problem in which a machine-learning model learns a mapping from the reduced calorimetric feature vector $\mathbf{x}^{(i)} \in \mathbb{R}^F$ to the event-level calibration factor $\alpha_{\mathrm{calib}}^{(i)}$:

$$\alpha_{\mathrm{calib}}^{(i)} = f_{\mathrm{ML}}(\mathbf{x}^{(i)}). \tag{20}$$

The choice of the regression model directly affects the calibration accuracy, statistical behavior, CPU-based prediction time, and suitability for further trigger-oriented latency studies. For this reason, the regression methods considered in this work were selected to evaluate different trade-offs between nonlinear modeling capability, variance reduction, residual-error correction, and computational cost.

All models are trained globally, using the full training sample without prior $(|\eta|, E_T)$ binning. The label `no-Bin` is therefore used to indicate that a single regressor is trained over the complete feature space, rather than fitting separate models in disjoint kinematic regions. After training, however, the models are evaluated both globally and within distinct $(|\eta|, E_T)$ regions. This strategy allows the stability of the global calibration function to be assessed across different kinematic regimes while avoiding the statistical fragmentation and boundary effects associated with bin-specific training.

The baseline tree-based methods considered in this study are Decision Trees, Gradient Boosted Decision Trees, and Random Forests. Decision Trees recursively partition the feature space using hierarchical decision rules and can capture nonlinear interactions, although they are prone to overfitting if not properly regularized [22]. Gradient Boosted Decision Trees build an additive ensemble of shallow trees trained sequentially to reduce prediction errors, making them effective for modeling structured nonlinear trends in the data [22,24,39]. Random Forests train an ensemble of decorrelated trees on bootstrap samples with random feature subsampling, and combine their predictions by averaging [22,23]:

$$f_{\mathrm{RF},\alpha}(\mathbf{x}^{(i)}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}^{(i)}), \tag{21}$$

where T_b denotes the bth regression tree. This averaging reduces prediction variance relative to a single tree while preserving the ability to model nonlinear dependencies among the calorimetric features.

In addition to these standard regressors, three hybrid ensemble strategies are evaluated to test whether complementary tree-based learning mechanisms can improve energy calibration without requiring explicit $(|\eta|, E_T)$ binning. Two of them are sequential residual-correction models, whereas the third one is based on stacked generalization.

The first residual-correction model, denoted `BDTRF no-Bin`, uses a BDT as the primary regressor and a Random Forest to model the remaining residuals. The first-stage prediction is

$$\alpha^{(i)}_{\mathrm{BDT}} = f_{\mathrm{BDT},\alpha}(\mathbf{x}^{(i)}), \tag{22}$$

with residuals

$$r^{(i)}_{\mathrm{BDT}} = \alpha^{(i)}_{\mathrm{true}} - \alpha^{(i)}_{\mathrm{BDT}}. \tag{23}$$

A Random Forest is then trained to predict $r^{(i)}_{\mathrm{BDT}}$, yielding the final estimator

$$\alpha^{(i)}_{\mathrm{calib}} = f_{\mathrm{BDT},\alpha}(\mathbf{x}^{(i)}) + f_{\mathrm{RF},r}(\mathbf{x}^{(i)}). \tag{24}$$

This configuration tests whether a variance-reducing ensemble can improve the residual structure left by the boosted model.

The second residual-correction model, denoted `RFBDT no-Bin`, reverses the order of the learners. A Random Forest is first trained to provide a stable baseline prediction,

$$\alpha^{(i)}_{\mathrm{RF}} = f_{\mathrm{RF},\alpha}(\mathbf{x}^{(i)}), \tag{25}$$

and a BDT is subsequently trained on the corresponding residuals,

$$r^{(i)}_{\mathrm{RF}} = \alpha^{(i)}_{\mathrm{true}} - \alpha^{(i)}_{\mathrm{RF}}. \tag{26}$$

The final prediction is given by

$$\alpha^{(i)}_{\mathrm{calib}} = f_{\mathrm{RF},\alpha}(\mathbf{x}^{(i)}) + f_{\mathrm{BDT},r}(\mathbf{x}^{(i)}). \tag{27}$$

This configuration evaluates whether a boosted residual model can refine the systematic errors remaining after a variance-stabilized Random Forest baseline.

The third hybrid strategy, denoted `RFBDTS no-Bin`, follows a stacked-generalization approach [22,25,26]. In this case, Random Forest and BDT regressors are trained in parallel, and their predictions are combined by a Ridge linear meta-learner:

$$\alpha^{(i)}_{\mathrm{calib}} = w_0 + w_1 f_{\mathrm{RF},\alpha}(\mathbf{x}^{(i)}) + w_2 f_{\mathrm{BDT},\alpha}(\mathbf{x}^{(i)}), \tag{28}$$

Table 2
Number of events in each $(|\eta|, E_T)$ region used for the bin-wise performance evaluation.

E_T interval (GeV)	$\|\eta\| \in [0, 0.43]$	[0.43, 0.83]	[0.83, 1.46]
[0, 44.7]	5462	6773	15 347
[44.7, 71.5]	6310	7193	12 677
[71.5, 103.5]	8160	8927	14 772
[103.5, 164.3]	15 305	15 642	7955
[164.3, 499.9]	13 493	7769	5954

where w_0, w_1, and w_2 are estimated under L_2-regularization. This formulation allows the meta-learner to exploit complementary predictive information from the two ensemble families while reducing the risk of overfitting through regularized weighting.

All Gradient Boosted Decision Tree models evaluated in this work were implemented using the LightGBM library. Comparative studies indicate that the relative performance of modern boosted-tree implementations, including XGBoost, LightGBM, and CatBoost, depends on the dataset characteristics, hyperparameter-optimization procedure, evaluation metric, and computational setting [40,41]. Accordingly, LightGBM was selected as the common boosted-tree implementation for the controlled comparison of the proposed calibration architectures.

Overall, the selected regressors provide a controlled comparison between single tree models, variance-reducing ensembles, boosting-based nonlinear regressors, residual-learning architectures, and stacked ensembles. This design enables the identification of calibration strategies that are not only accurate but also compact and computationally feasible for fast energy-reconstruction pipelines.

2.6. Binning, training and hyperparameter selection

The detector response in calorimetric systems is not uniform over the full phase space, as it may depend on both the pseudorapidity $|\eta|$ and the transverse energy E_T. For this reason, segmentation in $(|\eta|, E_T)$ regions is commonly used in high-energy physics analyses to inspect spatial and energy-dependent variations in reconstruction performance [19,42].

In this work, binning is used for performance evaluation rather than for model training. All regression models are trained globally, using the full training sample without prior $(|\eta|, E_T)$ segmentation. The bin-wise analysis is then performed a posteriori to assess whether the globally trained calibration models remain stable across different detector and energy regions. This strategy avoids the statistical fragmentation and boundary effects that may arise when separate models are trained independently in each bin, while still allowing localized performance differences to be identified.

Table 2 summarizes the $(|\eta|, E_T)$ binning adopted in this study and the corresponding number of events in each region.

The experimental design combines four feature-selection or feature-extraction strategies, namely `CORR`, `MUTI`, `PCA`, and `ICA`, each evaluated with four retained feature/component counts, and six regression models: `DTREE`, `BDT`, `RFOR`, `BDTRF`, `RFBDT`, and `RFBDTS`. This results in 96 reduced-representation configurations. In addition, the full-ring representation, `Rings[1:100]`, and the downsampled representation, `Rings/2`, are included as reference input configurations, yielding 108 evaluated configurations in total. These configurations are compared using the performance-evaluation framework described in Section 2.7, which combines global error metrics, computational efficiency, bin-wise stability, energy resolution, turn-on behavior, and statistical validation.

The dataset was initially partitioned into 70% training data and 30% independent test data. Hyperparameters were selected exclusively within the training partition through a nested cross-validation procedure with five outer folds and five inner folds. For each outer split, the inner loop evaluated a predefined grid search over the candidate

hyperparameter combinations and selected the configuration with the best validation performance. The outer loop was used to estimate the stability and generalization performance of the selected configurations within the training partition.

After hyperparameter selection, the final models were fitted using training data only and evaluated on the independent test partition. All data-dependent feature-selection and feature-extraction procedures, including CORR, MUTI, PCA, and ICA, were fitted exclusively on the corresponding training data and subsequently applied to the held-out data. This procedure prevents information from the test partition from influencing either feature construction or hyperparameter selection.

2.7. Performance evaluation

The performance of each calibration strategy is evaluated from complementary perspectives: global energy-reconstruction accuracy, computational efficiency, localized stability across $(|\eta|, E_T)$ regions, energy resolution, trigger-level response, and statistical significance. This multi-criteria evaluation is intended to identify models that are accurate, stable across the evaluated benchmark regions, compact, and computationally efficient in the relative CPU-based comparison considered here.

2.7.1. Global energy-error metrics

The global calibration performance is quantified by comparing the calibrated energy E_{calib} with the true energy E_{true}. For each event, the calibration error is defined as

$$\epsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}, \tag{29}$$

where calib denotes the machine-learning model used to estimate the calibration factor. For the Standard FastCalo baseline, the corresponding error is

$$\epsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}. \tag{30}$$

An ideal calibration method would yield errors centered around zero and with minimal dispersion. Therefore, two complementary metrics are used:

- Root Mean Squared Error (RMSE), which measures the average magnitude of the error and penalizes large deviations;
- Interquartile Range (IQR), which quantifies the central dispersion of the error distribution and is robust to outliers and non-Gaussian tails.

To summarize the joint behavior of these two error-based metrics, we define

$$d_0 = \sqrt{\text{RMSE}^2 + \text{IQR}^2}. \tag{31}$$

Lower values of d_0 indicate better simultaneous performance in terms of global accuracy and error dispersion. This score is restricted to predictive performance and does not include computational cost or model compactness.

2.7.2. Computational efficiency and global ranking

Since the intended application involves high-throughput trigger-level environments, predictive performance must be assessed together with computational efficiency. Two additional quantities are therefore considered:

- Number of features, F, which reflects the dimensionality of the input representation used during inference;
- Prediction time per event, T, which measures the time required by the trained model to estimate the calibration factor after the input features are available.

Training time is not included in this evaluation because training is performed offline and does not affect real-time operation. The Standard FastCalo baseline is treated as a reference with $F_{\text{fast}} = 1$ and negligible additional inference latency, since its energy estimate is directly available from the fast reconstruction chain.

To combine predictive accuracy, compactness, and trained-model prediction time into a single global score, we define:

$$D_0 = \sqrt{\text{IQR}^2_{(N)} + \text{RMSE}^2_{(N)} + F^2_{(N)} + T^2_{(N)}}, \tag{32}$$

where $\text{IQR}_{(N)}$, $\text{RMSE}_{(N)}$, $F_{(N)}$, and $T_{(N)}$ are min–max normalized values computed across all evaluated configurations. Normalization is required because the four quantities have different scales and physical units.

Equal weights are assigned to all four terms in order to define D_0 as a transparent baseline aggregator, avoiding subjective prioritization of either predictive accuracy or computational efficiency. Thus, lower values of D_0 indicate better overall trade-offs. The comparison between d_0 and D_0 is also used to assess whether the ranking of the best-performing configurations is driven mainly by predictive quality or by efficiency-related gains. The D_0 score is used only as a transparent equal-weight ranking criterion, not as a universal deployment metric. For this reason, RMSE, IQR, number of features, and inference time are also reported separately, allowing different operational priorities to be considered. Likewise, the measured inference times should be interpreted as relative benchmarks on conventional hardware, while dedicated validation would still be required for production trigger environments.

Inference times were measured using the following hardware configuration: Intel® Core™ i7-7500U CPU, 20 GB DDR4 RAM, 1 TB SSD, and NVIDIA® GeForce® 940MX GPU with 2 GB of memory.

Although this setup is modest compared with the infrastructure used in large-scale High-Level Trigger systems [43–45], it provides a relative CPU-based benchmark for comparing the inference cost of the evaluated models. The reported times correspond only to the prediction stage after feature availability and do not include online feature construction, memory-access or transfer overheads, model-loading costs, batching effects, fixed-point quantization, hardware-specific implementation, or worst-case end-to-end trigger latency. They should therefore not be interpreted as direct deployment-latency estimates for a production trigger system.

2.7.3. Bin-wise calibration-factor evaluation

After the global ranking, the top-performing configurations are evaluated locally within the $(|\eta|, E_T)$ regions defined in Table 2. This bin-wise analysis is performed to verify whether the globally trained models remain stable across different detector and energy regions.

The predicted calibration factor α_{calib} is compared with the target calibration factor α_{true}, defined in Eq. (9). For each bin, three metrics are computed: IQR, Mean Absolute Error (MAE), and RMSE. The IQR quantifies the dispersion of the predicted calibration factors, whereas MAE and RMSE measure the deviation from the ideal multiplicative correction.

This localized evaluation is important because global metrics may hide instabilities or systematic deviations in specific regions of the detector acceptance. Therefore, the bin-wise analysis complements the global ranking by assessing the regional consistency of the selected calibration strategies.

2.7.4. Energy-resolution evaluation

Energy resolution is used as an additional performance indicator to characterize the precision of the calibrated energy estimates over the full energy range [16,32]. For each event, the calibrated response is defined as

$$r = \frac{E_{\text{calib}}}{E_{\text{true}}}, \tag{33}$$

with the corresponding relative residual

$$\delta = \frac{E_{\text{calib}} - E_{\text{true}}}{E_{\text{true}}} = r - 1. \tag{34}$$

The energy resolution is estimated from the dispersion of r, or equivalently δ, within each energy interval. When the response distribution is approximately Gaussian, this dispersion can be represented by the standard deviation, such that

$$\frac{\sigma_E}{E} \approx \sigma_r \approx \sigma_\delta. \tag{35}$$

Because calorimetric residuals may exhibit non-Gaussian tails or asymmetries, a robust quantile-based estimator is also considered. The interquartile range is defined as

$$\text{IQR}(X) = Q_{75}(X) - Q_{25}(X), \tag{36}$$

and may be converted into a Gaussian-equivalent width through

$$\sigma_{\text{eq}} \approx \frac{\text{IQR}}{1.349}. \tag{37}$$

In this work, the energy resolution is evaluated as a function of energy by grouping events into energy intervals and computing the dispersion of the calibrated response in each interval. This analysis complements the global RMSE and IQR metrics by showing how the calibration quality evolves across the energy range.

2.7.5. Turn-On curve evaluation

The operational impact of the calibration models on trigger-level event selection is evaluated through turn-on curves derived from the calibrated transverse energy [30,32,33]. This analysis quantifies how improvements in energy calibration affect the sharpness and stability of the selection around a nominal energy threshold.

For a given threshold E_{thr}, the offset variable is defined as

$$\Delta\lambda = E_{\text{calib}} - E_{\text{thr}}. \tag{38}$$

The turn-on curve is then obtained as the event acceptance probability as a function of $\Delta\lambda$, i.e., the fraction of events passing the selection criterion after calibration.

A sharper turn-on curve indicates reduced migration of events across the threshold and therefore a more stable trigger response. In addition, the difference between the ML-based and Standard FastCalo acceptance curves is evaluated as a function of $\Delta\lambda$, providing a direct measure of the selection-efficiency gain induced by the proposed calibration models.

2.7.6. Statistical validation

A statistical validation is performed to determine whether the observed improvements over the Standard FastCalo baseline are significant and not attributable to random fluctuations. Since the calibrated and uncalibrated estimates are obtained from the same physical events, all comparisons are treated as paired tests.

The paired error difference is defined as

$$d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}. \tag{39}$$

For the mean error, the directional hypothesis is

$$H_0 : \mu_d \geq 0 \quad \text{vs.} \quad H_1 : \mu_d < 0, \tag{40}$$

where μ_d is the mean of the paired differences. A large-sample paired z-test is used as the primary test, supported by a paired Student's t-test and a Wilcoxon signed-rank test as robustness checks [46–51].

For the error variance, the directional hypothesis is

$$H_0 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} \geq 1 \quad \text{vs.} \quad H_1 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} < 1. \tag{41}$$

The Pitman–Morgan test is used as the primary paired variance test, complemented by paired bootstrap confidence intervals for the variance ratio and by the classical F-test as a reference comparison [46,52–55].

Normality is assessed using the D'Agostino–Pearson K^2 test and the Anderson–Darling test, complemented by Q–Q plots [56–61]. The Shapiro–Wilk test is not used for samples larger than $n = 5000$, due to its known limitations in very large datasets [59]. All tests are performed at a significance level of $\alpha = 0.05$. For each comparison, the test statistic, p-value, and one-sided 95% confidence interval are reported.

The statistical comparisons in this work are performed only between each selected calibration model and the Standard FastCalo baseline; they are not intended to rank the hybrid models against one another through hypothesis testing. For the event-wise error $\varepsilon = E_{\text{true}} - E_{\text{calib}}$, the paired tests assess whether the calibrated estimates exhibit an error mean closer to zero and a lower error variance than the corresponding Standard FastCalo estimates. The paired statistical analysis is therefore sufficient for its specific purpose of testing improvement in the central tendency and variance of the energy-estimation error relative to the baseline. Because the sample size is large, statistical significance alone is not interpreted as evidence of practical relevance. The practical importance of the observed differences is assessed jointly through the magnitude of the reductions in RMSE, IQR, energy resolution, and threshold offset, together with the bin-wise stability analyses across $(|\eta|, E_T)$ regions. Additional diagnostics, such as upper-tail residual percentiles, worst-bin summaries, and maximum regional bias, may further characterize localized or rare-event behavior, but are complementary to the predefined statistical and performance-evaluation protocol adopted in this work.

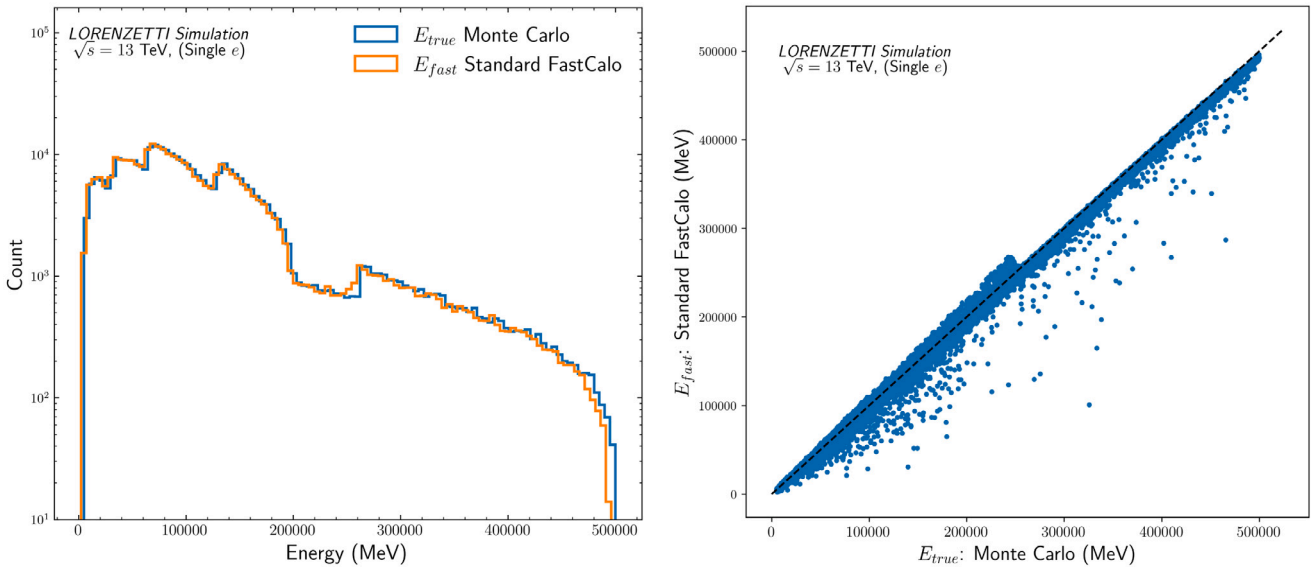
3. Results

Fig. 5(a) presents histograms of True Energy (E_{true}) and Standard FastCalo Energy (E_{fast}), both showing similar distributions. However, discrepancies are evident in the scatter plot depicted in Fig. 5(b), where distortions in the mean response and variance are apparent. Specifically, mean distortions exhibit a discontinuity around 250 GeV, while variance distortions are asymmetric, with greater dispersion observed below the $E_{\text{true}} = E_{\text{fast}}$ axis.

The discontinuity around 250 GeV is part of the controlled fast-response benchmark used in this work. It was introduced to emulate a localized non-smooth response distortion and to test whether the proposed calibration models can correct nonlinear effects beyond smooth global trends. It is therefore used as a robustness stress test, rather than as a claim of reproducing a specific detector artifact.

The same step-like perturbation and dataset were previously adopted in a Lorenzetti-based calibration study [21]. They were retained here to preserve a consistent and reproducible benchmark for assessing methodological developments under identical simulated response conditions. In the present work, the benchmark is extended through hybrid tree-based regressors, fixed-cardinality feature-selection and feature-extraction strategies, and a broader measurement-oriented evaluation protocol. This perturbation should not be interpreted as a detector-specific artifact or as a complete model of realistic calorimeter-response imperfections. Instead, it provides a controlled nonlinear and non-smooth response stress test under which all calibration models and the Standard FastCalo baseline are evaluated. Since such a perturbation may interact differently with different model families, including tree-based models, the reported improvements should be interpreted as relative performance gains within this controlled benchmark. A dedicated comparison using otherwise identical simulations with and without the perturbation would be required to isolate its effect explicitly and is left for future work.

Further analysis in Fig. 6(a) examines the relationships between E_{true} and E_{fast}, including the calibration error $E_{\text{true}} - E_{\text{fast}}$, normalized calibration error $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$, and calibration factor $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$. All plots are normalized using the z-score for better comparison. Z-score is defined as the number of standard deviations a data point is from the mean [46].



(a) (b)

Fig. 5. Comparison of true and fast reconstructed energies: (a) energy histogram comparing the true transverse energy E_{true} and the Standard FastCalo energy E_{fast} generated with the Lorenzetti simulator over the 0–500 GeV range; (b) scatter plot of E_{true} versus E_{fast}. The scatter plot shows an overall correlation between the two quantities, but also exhibits a visible step-like nonlinear distortion around 250 GeV, together with changes in the mean response and variance at higher energies. This distortion was intentionally introduced in the simulated fast-energy response to provide a controlled robustness test for the proposed calibration methods against nonlinear and non-smooth detector-response effects.

Fig. 6(b) shows that the residual distribution $E_{\text{true}} - E_{\text{fast}}$ closely follows a Student's t-distribution, in contrast to the more skewed behaviors observed for the relative errors $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$ and the energy ratio $E_{\text{true}}/E_{\text{fast}}$, both of which deviate significantly from normality. These results indicate that the absolute error $E_{\text{true}} - E_{\text{fast}}$ is a more appropriate metric for statistical inference. Consequently, it was adopted in the t and F-tests used for the validation of calibration accuracy and variance.

The calibration factor α_{true}, crucial for ML methods estimating α_{calib}, is examined in Fig. 7. The boxplots across different E_T and $|\eta|$ bins reveal that estimated calibration factors should ideally match α_{true} to minimize calibration errors ($\epsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$).

Fig. 8 presents the distribution of normalized energy depositions across the 100 concentric rings generated by the *Ringer* algorithm for the full set of simulated events. Each box represents the interquartile range (IQR) of the normalized energy levels in a given ring, obtained via min–max normalization applied independently to each event.

The vertical dashed lines indicate the approximate boundaries corresponding to the different longitudinal layers of the calorimeter: the Pre-Sampler (PS), the Electromagnetic layers (EM1, EM2, EM3), and the Hadronic layers (HAD). As expected for electromagnetic showers initiated by electrons, the largest energy concentrations occur in the EM2 region, followed by smaller but non-negligible contributions in the PS and EM1 regions. Minimal energy is deposited in the EM3 and hadronic layers, consistent with the containment of the electromagnetic shower within the electromagnetic calorimeter.

This visualization highlights the characteristic energy deposition pattern across calorimeter layers, and illustrates how the *Ringer* algorithm captures radial information effectively in a compact 100-dimensional feature space. These transformed features serve as the foundation for the subsequent analysis, where we evaluate various methods of feature selection and extraction to identify the most informative representations for energy calibration.

3.1. Feature selection and extraction evaluation

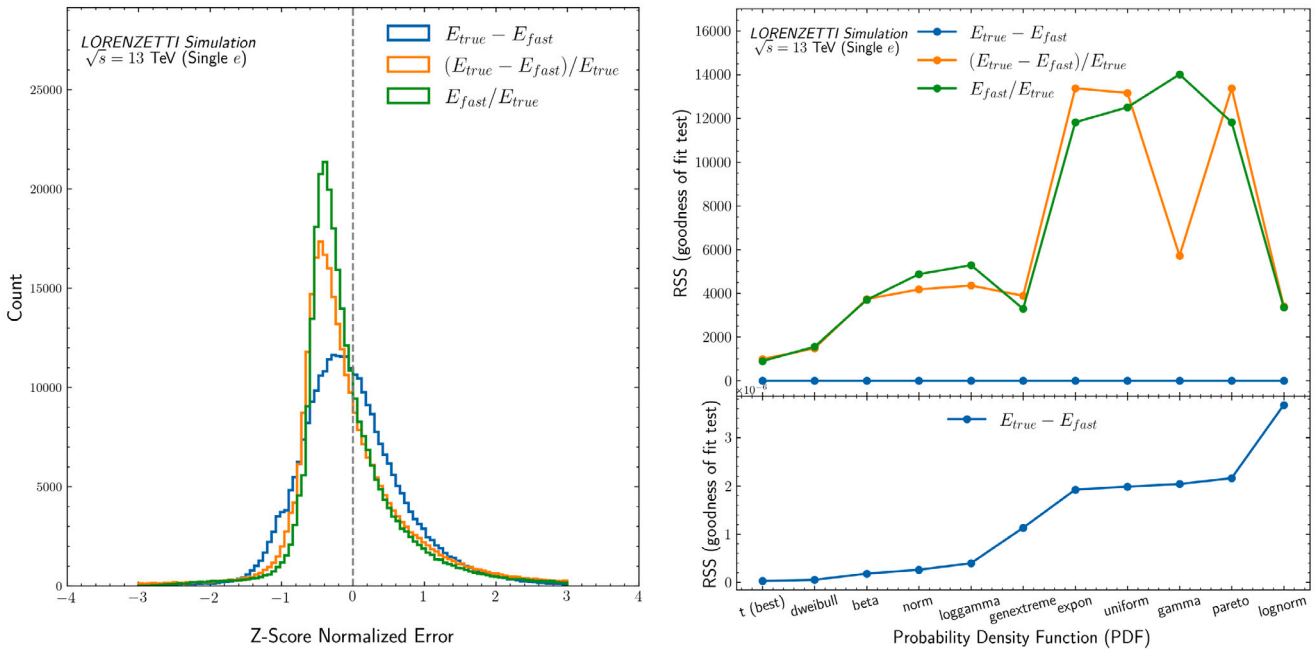
Feature selection and extraction methods are analyzed through heatmaps shown in Fig. 9. These maps depict the correlation, mutual information, PCA, and ICA relationships with E_{true}. High-correlation regions depicted in dark blue are guiding the selection of attributes for ML methods.

Table 3
Selected/extracted features and number of features (F) by correlation, mutual information, PCA and ICA approaches.

Selection/Extraction	Features	F
Rings[1:100]	Rings[1:100]	100
Rings/2	Rings[15, 9:40, 73:76, 81:84, 89:94]	50
CORR-50	Rings[1, 9:42, 73:85, 93, 97]	50
CORR-10	Rings[9:11, 73:78, 82]	10
CORR-5	Rings[73:77]	5
CORR-2	Rings[75, 76]	2
MUTI-50	Rings[1, 9:39, 71:85, 89, 93, 97, 98]	50
MUTI-10	Rings[9:13, 73:77]	10
MUTI-5	Rings[9, 73:76]	5
MUTI-2	Rings[74, 75]	2
PCA-50	PCA[1:50]	50
PCA-10	PCA[1:10]	10
PCA-5	PCA[1:5]	5
PCA-2	PCA[1:2]	2
ICA-50	ICA[1:50]	50
ICA-10	ICA[1:10]	10
ICA-5	ICA[1:5]	5
ICA-2	ICA[1:2]	2

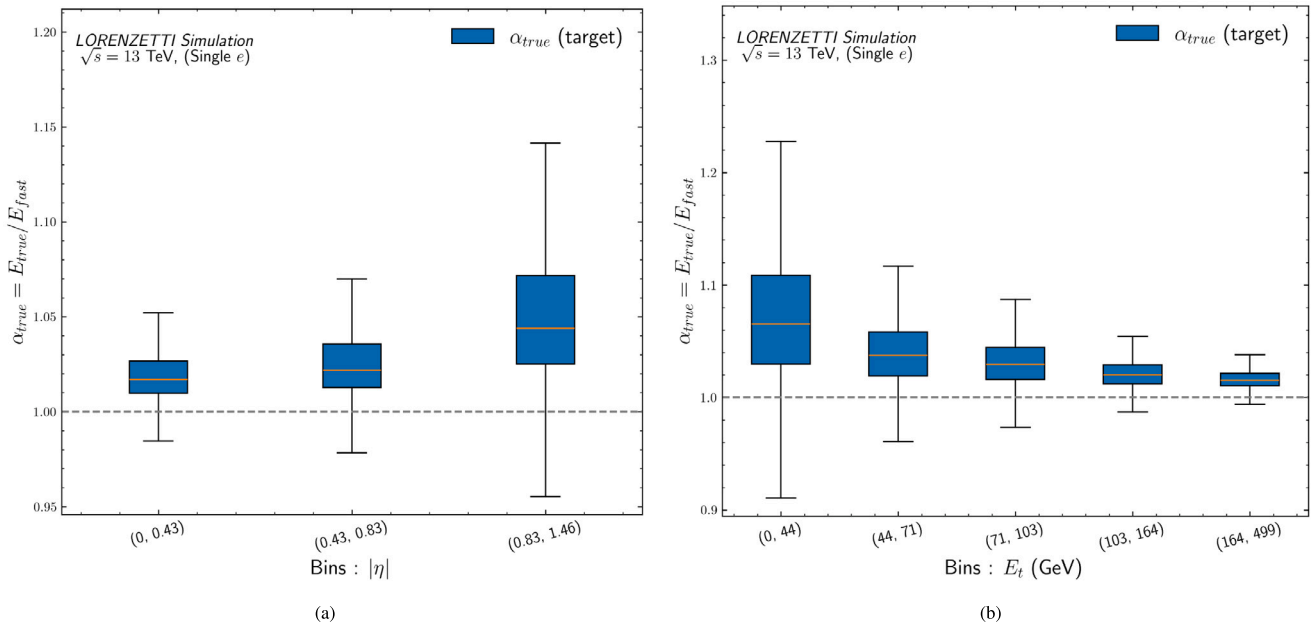
Table 3 summarizes the sets of features selected or extracted by each dimensionality reduction method evaluated in this study. The first row (`Rings[1:100]`) represents the original input space, composed of all 100 normalized ring energies generated by the Lorenzetti Showers framework. The second row (`Rings/2`) denotes a reduced feature space in which the ring variables from each calorimeter region (PS, EM1–3, and HAD) are downsampled by a factor of two.

The subsequent rows summarize the feature subsets selected by the correlation-based (CORR) and mutual information based (MUTI) strategies at different selection levels. For instance, `CORR-50` retains the 50 rings exhibiting the largest absolute Pearson correlation coefficients with the true energy. As the number of retained features is reduced



(a) (b)

Fig. 6. Comparison of energy errors and goodness of fit tests of errors: (a) Z-score normalized relationships between E_{true} and E_{fast} illustrating energy errors. The legend presents different methods of evaluating the error: the absolute difference ($E_{\text{true}} - E_{\text{fast}}$), the relative difference normalized by the true energy $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$, and the ratio $E_{\text{fast}}/E_{\text{true}}$. The objective is to identify the relation whose distribution is closest to a Gaussian, thereby simplifying the treatment of calibration errors; (b) Residual Sum of Squares (RSS) goodness-of-fit tests were performed to evaluate different formulations of the calibration error between E_{true} and E_{fast}, aiming to determine the most statistically appropriate distribution for each case. Among the evaluated formulations, the absolute error ($E_{\text{true}} - E_{\text{fast}}$) yielded the lowest RSS, indicating the best overall fit. The blue curve, corresponding to this formulation, presented significantly smaller residuals compared to the green and orange curves, appearing nearly flat in the upper plot. A magnified view of the blue curve is shown in the lower panel. The results suggest that the Student's t-distribution provides the best fit for modeling the calibration error.



(a) (b)

Fig. 7. Boxplots of the true calibration factor α_{true} across (a) the pseudorapidity $|\eta|$, and (b) transverse energy E_T bins. The boxplots illustrate the distribution of calibration factors within each bin, where the central line represents the median, the box denotes the interquartile range (IQR), and the whiskers extend to 1.5 times the IQR from the lower and upper quartiles. This visualization highlights the uncertainty and variability of the calibration factors per bin, which serve as the target for the ML calibration methods.

(e.g., `CORR-2`), the input representation becomes increasingly compact, retaining only the two most strongly correlated ring components (e.g., `Rings[75, 76]`).

Similarly, the `MUTI`-based method selects features according to their mutual information with the true energy. As the retained cardinality m decreases, the representation becomes increasingly com-

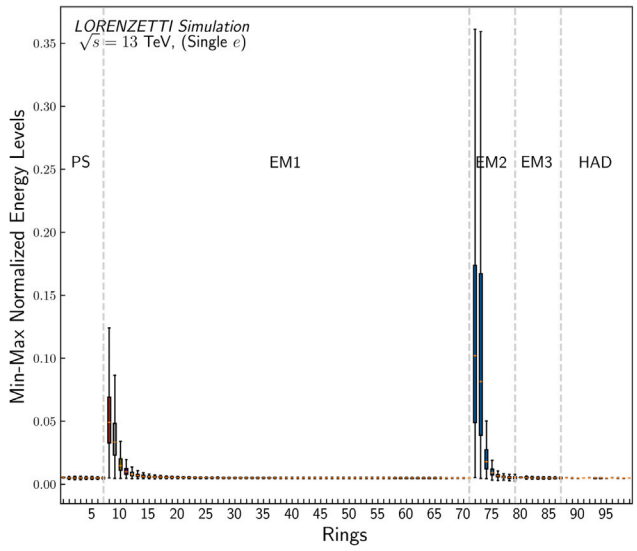


Fig. 8. Boxplot of min–max normalized energy levels across the 100 concentric rings defined by the *Ringer* algorithm for 300,000 simulated single-electron events. Each box represents the distribution of normalized energy values in one ring across all events. Vertical dashed lines delineate the approximate positions of the calorimeter layers: PreSampler (PS), Electromagnetic Layers (EM1–EM3), and Hadronic Layers (HAD). The concentration of energy in EM2 is consistent with typical electromagnetic shower development.

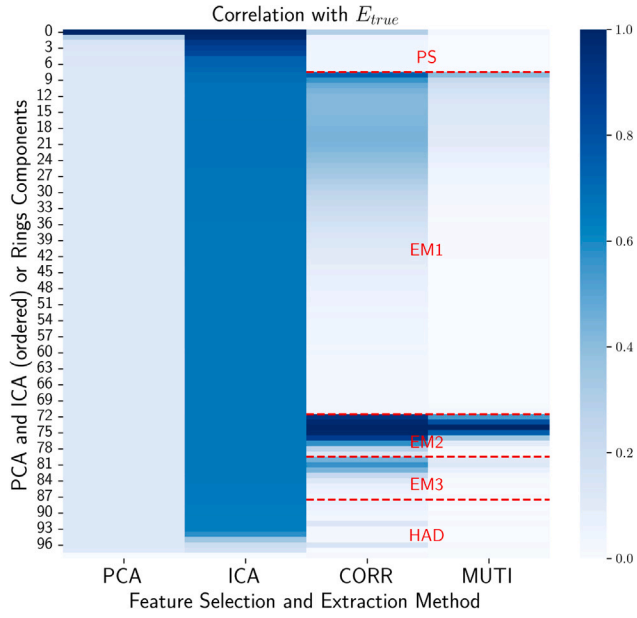


Fig. 9. Heatmap of the relationship between the 100 input components and the true energy E_{true} for four feature selection/extraction strategies: PCA, ICA, CORR, and MUTI. Each column corresponds to one method, and each row to a component index. Darker blue shades indicate stronger relevance to E_{true}. The vertical red dashed lines indicate the approximate calorimeter-region boundaries (PS, EM1–EM3, and HAD), which are physically meaningful only for CORR and MUTI because they operate directly on the original ring energies. PCA and ICA components are transformed latent variables and therefore do not map directly to specific calorimeter regions. For both PCA and ICA, more than 99% of the total variance is captured by the first 10 components.

pact, with `MUTI-2` retaining only the two most informative rings, `Rings[74,75]`. It is noteworthy that the strong correlation and mutual information observed for ring variables 72–78 can be attributed to their spatial arrangement within the calorimeter layers. These rings are positioned close to each other, leading to similar responses to incident particles due to shared energy deposition characteristics. This spatial relationship increases the likelihood of correlated measurements, as they capture similar energy distributions from particle showers.

For the PCA and ICA methods, which perform feature extraction rather than selection, the table lists the top $m \in \{50, 10, 5, 2\}$ components ranked by their absolute correlation with the true energy. For example, `PCA-5` retains the first five principal components (`PCA[1:5]`), capturing the directions of highest variance aligned with the energy prediction task.

Table 3 highlights how different dimensionality-reduction methods reduce the feature space from 100 original inputs to as few as two features or components, enabling significant gains in inference speed and storage efficiency.

It is also important to interpret the compact PCA and ICA-based representations in the context of the simulated detector conditions. These representations are not used here as explicit pile-up-suppression mechanisms, but as dimensionality-reduction strategies for the ring-based input space. This reduction may be useful in high-throughput calorimetric systems because storing or processing a small number of latent components can reduce storage requirements and inference cost when compared with the full 100-ring representation. Accordingly, the compact representations should be interpreted as preserving part of the calibration-relevant shower information under the simulated conditions considered here, rather than as dedicated pile-up-mitigation methods.

3.2. Global energy-error evaluation

This section evaluates the calibration error obtained by combining the feature-selection and feature-extraction strategies `CORR`, `MUTI`, `PCA`, and `ICA` with the tree-based regression models `DTREE`, `BDT`, `RFOR`, `BDTRF`, `RFBDT`, and `RFBDTS`. The analysis is based on the event-wise calibration error,

$$\epsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}, \tag{42}$$

and compares the predictive accuracy and dispersion achieved by each configuration.

Fig. 10 shows the calibration-error distributions and the corresponding boxplots for each feature-reduction strategy. In the boxplots, the central line represents the median error, the left and right box edges correspond to the first and third quartiles, and the box width therefore represents the interquartile range (IQR). The whiskers extend to the most extreme points within 1.5 times the IQR, while the black markers indicate the RMSE. This representation allows the bias, central dispersion, tail behavior, and large-error sensitivity of each configuration to be compared simultaneously.

Across all feature-reduction strategies, the Standard FastCalo baseline exhibits the largest deviations from zero and the broadest error distributions, confirming that the uncalibrated fast energy estimate provides a conservative reference for comparison. The proposed ML-based calibration strategies reduce the calibration-error dispersion with respect to this baseline, although the magnitude of the improvement depends on both the feature representation and the regression model.

For correlation-based feature selection, the best performing configuration is `RFBDT no-Bin: CORR-50`, which achieves the lowest IQR and RMSE among the `CORR` configurations. Relative to Standard FastCalo, this model reduces the IQR by 36.18% and the RMSE by 53.40%. A more compact alternative, `RFBDTS no-Bin: CORR-10`, also remains competitive despite using only 10 features, reducing the IQR and RMSE by 28.05% and 47.10%, respectively. This indicates that correlation-based feature selection can preserve a substantial fraction of the relevant calorimetric information even under strong dimensionality reduction.

A similar behavior is observed for mutual-information-based feature selection. The best result in this group is obtained with `RFBDT no-Bin: MUTI-50`, which reduces the IQR by 34.15% and the RMSE

by 58.44% relative to Standard FastCalo. The compact configuration RFBDTS no-Bin: MUTI-10 also provides relevant gains, with reductions of 25.61% in IQR and 44.08% in RMSE while using a feature set five times smaller than the 50-feature configuration. These results suggest that mutual information is effective in identifying informative calorimetric rings, but that the highest accuracy still benefits from retaining a larger subset of selected features.

For PCA-based feature extraction, the best configuration is BDT no-Bin: PCA-10, which achieves reductions of 27.64% in IQR and 51.39% in RMSE compared with Standard FastCalo. The configuration BDT no-Bin: PCA-5 provides a more compact alternative, using only half the number of principal components while still reducing the IQR by 25.61% and the RMSE by 49.37%. This small performance loss indicates that most of the useful calibration information captured by PCA is concentrated in a relatively low-dimensional subspace.

For ICA-based feature extraction, the best performance is obtained with RFBDTS no-Bin: ICA-5, which reduces the IQR by 24.80% and the RMSE by 51.64% relative to Standard FastCalo. The configuration BDT no-Bin: ICA-5 also shows competitive behavior, with reductions of 25.61% in IQR and 51.64% in RMSE. The proximity between these results suggests that ICA-based representations with five components are sufficiently compact to retain the main independent structures relevant to the calibration task, although their overall dispersion reduction is slightly weaker than that achieved by the best CORR and MUTI configurations.

These results show that the largest RMSE reductions are achieved by hybrid or boosted tree-based configurations, particularly RFBDT no-Bin: MUTI-50, RFBDT no-Bin: CORR-50, and compact PCA/ICA-based BDT configurations. From an operational perspective, the comparison also highlights an important trade-off: configurations with 50 selected features generally provide the strongest error reduction, whereas compact representations such as CORR-10, MUTI-10, PCA-5, and ICA-5 retain competitive performance with substantially lower input dimensionality. This trade-off motivates the subsequent global ranking based on (d_0) and (D_0), where predictive accuracy, error dispersion, model compactness, and inference latency are jointly considered.

3.2.1. Global ranking and accuracy–efficiency trade-off

To compare the calibration strategies under both predictive and operational criteria, all evaluated configurations were ranked according to the two composite metrics defined in Section 2.7. The first metric, d_0, combines only RMSE and IQR and therefore ranks the methods according to calibration accuracy and error dispersion. The second metric, D_0, extends this ranking by incorporating the number of input features F and the inference time per event T, thereby accounting for compactness and latency constraints.

The comparison includes both the full-ring representation, Rings[1:100], and the reduced feature representations obtained through CORR, MUTI, PCA, and ICA.

This design allows the ranking to distinguish between configurations that maximize absolute calibration performance and those that provide more favorable accuracy–efficiency trade-offs.

Fig. 11 summarizes the top-ranked configurations under both criteria. Panels (a) and (b) show the 10 best configurations according to d_0 and D_0, respectively, expressed as percentage improvements relative to the Standard FastCalo baseline. Panels (c) and (d) show the corresponding calibration-error distributions and boxplots for the methods ranked by d_0 and D_0. In these boxplots, the IQR represents the central dispersion of the residual distribution, while the black markers indicate the RMSE.

Table 4 reports the four configurations that best summarize the two operating regimes identified by the ranking analysis. The full-ring models, RFBDT no-Bin: Rings[1:100] and BDT no-Bin: Rings[1:100], achieve the best absolute calibration accuracy and therefore dominate the d_0-based ranking. In contrast, the compact latent-space configurations, BDT no-Bin: ICA-5 and BDT no-Bin: PCA-10, achieve the most favorable D_0 values because they preserve competitive calibration performance while using substantially fewer input features.

Relative to Standard FastCalo, all four ML-based configurations substantially reduce both the IQR and the RMSE of the calibration error. The best d_0 value is obtained by RFBDT no-Bin: Rings[1:100], which reduces d_0 from 4666.42 MeV to 2092.11 MeV, corresponding to an improvement of approximately 55.17%. This configuration also reduces the IQR by 43.39% and the RMSE by 60.61%, indicating simultaneous gains in robustness and average predictive accuracy. The closely related BDT no-Bin: Rings[1:100] configuration provides comparable performance, with reductions of 43.64% in IQR and 60.17% in RMSE.

The compact configurations exhibit a different but operationally relevant behavior. Although their absolute error reduction is smaller than that of the full-ring models, BDT no-Bin: PCA-10 and BDT no-Bin: ICA-5 still reduce the RMSE by more than 51% relative to Standard FastCalo, while using only 10 and 5 input components, respectively. This indicates that most of the calibration-relevant information can be retained in a low-dimensional latent representation. As a consequence, these configurations become preferable under the D_0 criterion, which explicitly penalizes larger input dimensionality and higher inference cost.

The prediction times correspond to the trained-model prediction stage of each configuration under the CPU-based benchmark described in Section 2.7.2. Thus, for single-regressor, residual-learning, and stacked models, the measured time includes all model-evaluation stages required to produce the final calibration factor. These measurements therefore provide a consistent relative comparison among the evaluated architectures within the same computational environment. Nevertheless, direct deployment in trigger-dedicated architectures would require additional implementation studies. In such systems, the end-to-end latency may also depend on feature-construction overhead, memory-access patterns, numerical representation, parallelization strategy, and worst-case throughput constraints of the target hardware. Therefore, the reported prediction times should be interpreted as relative CPU-based inference benchmarks, rather than as direct estimates of production-trigger latency.

The systematic deviations observed in the Standard FastCalo baseline are consistent with the nonlinear response patterns shown in Fig. 5(b) and with the broader residual distributions reported in Figs. 10(a)–10(d). These deviations arise from the combined effects of detector response, electronic noise, pile-up, and limitations of the fast reconstruction chain. The ML-based calibrators reduce these effects by learning event-level correction factors from the ring-based shower topology, thereby bringing E_{calib} closer to E_{true} and narrowing the residual distributions.

The ranking results also support the use of compact dimensionality-reduced representations. In particular, PCA and ICA concentrate most of the relevant information in the first few components, which explains why PCA-10 and ICA-5 remain competitive despite their reduced dimensionality. This behavior is especially relevant for future trigger-oriented studies, where calibration models would need to satisfy strict constraints on memory footprint, prediction time, and data movement. The selected dimensionality limits therefore represent a deliberate trade-off between model simplicity and calibration performance.

All four top-ranked configurations exhibit trained-model prediction times below 7 μs per event on the conventional CPU-based hardware used in this study. These measurements do not establish direct readiness for deployment in a production trigger chain, but they indicate that the proposed models are sufficiently lightweight to motivate further latency studies in dedicated online-processing environments [30]. Overall, the comparison shows that full-ring hybrid models provide the highest absolute calibration performance, whereas compact PCA- and ICA-based models offer the most favorable accuracy–efficiency compromise.

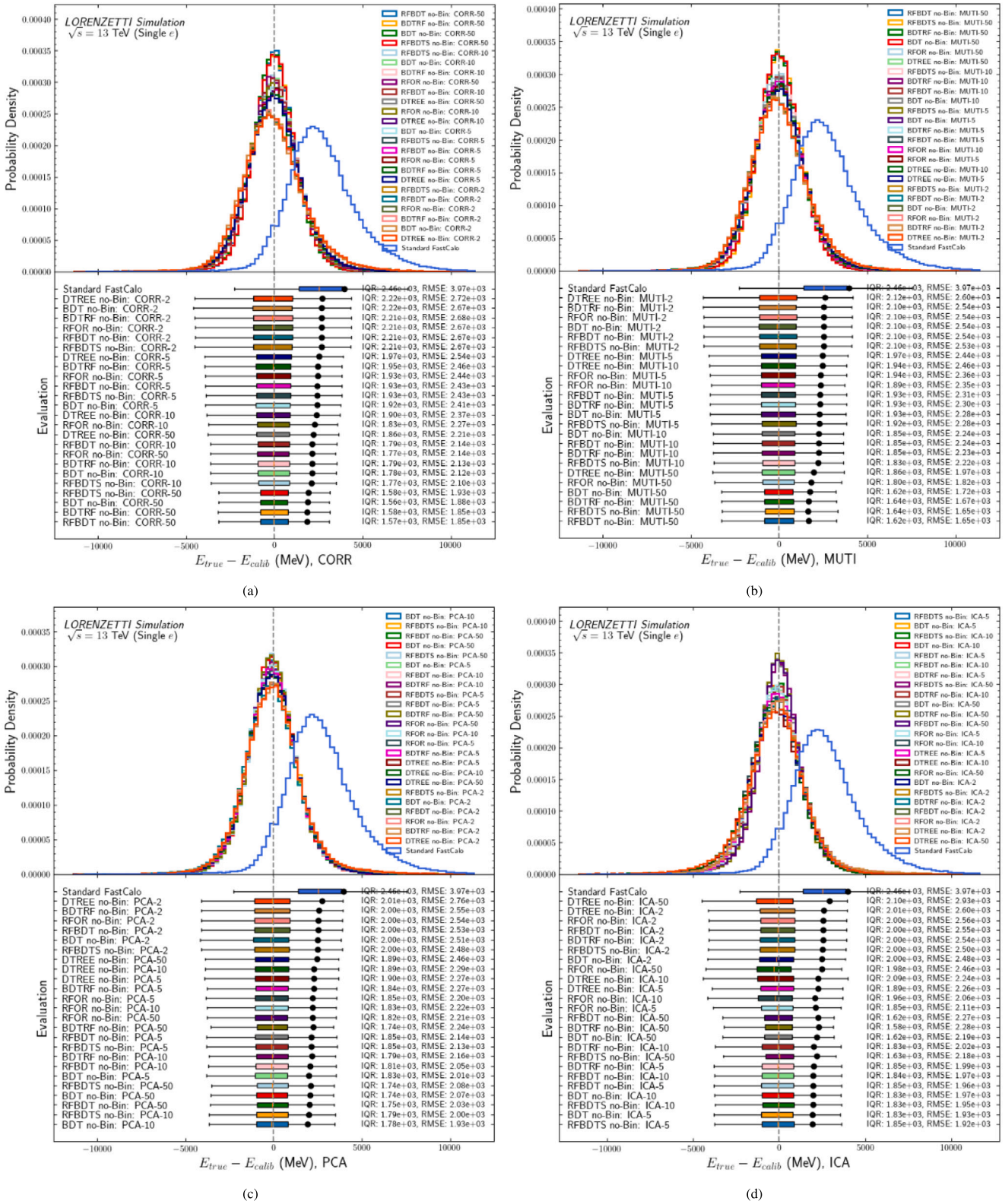


Fig. 10. Calibration-error distributions for $E_{\text{true}} - E_{\text{calib}}$ obtained with different feature-selection and feature-extraction strategies: (a) correlation-based selection (CORR), (b) mutual-information-based selection (MUTI), (c) principal component analysis (PCA), and (d) independent component analysis (ICA). The upper panels show the error histograms, while the lower panels show the corresponding boxplots. In each boxplot, the box represents the interquartile range, the central line indicates the median, and the black marker denotes the RMSE.

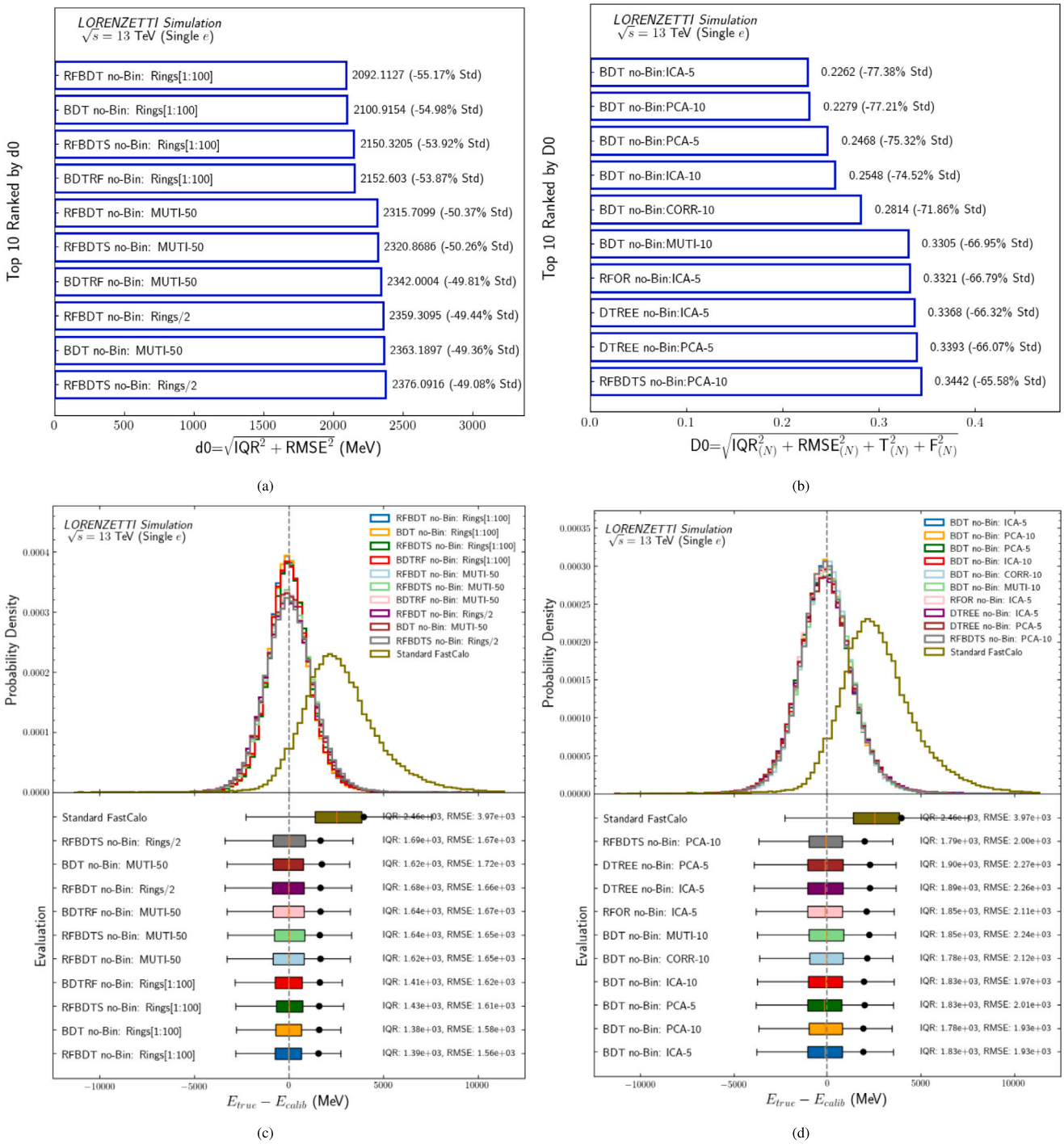


Fig. 11. Ranked-method comparison according to the d_0 and D_0 metrics. Panels (a) and (b) show the top-10 configurations ranked by d_0 and D_0, respectively, with percentage improvements relative to Standard FastCalo. Panels (c) and (d) show the corresponding distributions of the calibration error $E_{\text{true}} - E_{\text{calib}}$ and the associated boxplots. The IQR quantifies the central dispersion of the residuals, while the black markers indicate the RMSE.

3.2.2. *Bin-wise calibration-factor evaluation*

The bin-wise analysis evaluates whether the globally trained calibration models remain stable across different detector and energy regions. For this purpose, the calibration factors predicted by the top-ranked configurations are compared with the reference factor

$$\alpha_{\text{true}} = \frac{E_{\text{true}}}{E_{\text{fast}}}, \tag{43}$$

which represents the ideal multiplicative correction required to transform the Standard FastCalo estimate into the true transverse energy. The comparison is performed in the $(|\eta|, E_T)$ regions defined previously, using the IQR, MAE, and RMSE of the calibration-factor estimates.

Fig. 12 shows the IQR of the calibration-factor distributions as a function of $|\eta|$ and E_T. The IQR increases with $|\eta|$, indicating larger dispersion of the calibration factors away from the central detector region. This behavior is consistent with the increasing complexity of the calorimetric response at larger pseudorapidity values. Conversely, the IQR decreases as E_T increases, reflecting the improved signal-to-noise ratio and the reduced relative impact of fluctuations at higher transverse energies. Across the evaluated regions, the ML-based meth-

Table 4
Performance summary of the four top-ranked calibration configurations in terms of interquartile range (IQR), root mean squared error (RMSE), prediction time per event T, number of input features F, and the composite metrics d_0 and D_0. Results were obtained from the independent test set, corresponding to 30% of the full dataset. Standard FastCalo is used as the reference baseline for all relative comparisons. IQR and RMSE values are reported as mean and standard deviation obtained from a five-fold resampling procedure on the test set. The prediction time per event is given as the mean measured inference time.

Method	IQR (MeV)	RMSE (MeV)	IQR (% Std)	RMSE (% Std)	T (μs)	F	d_0 (MeV)	D_0
Standard FastCalo	2457.11 ± 9.89	3967.12 ± 6.74	–	–	–	–	4666.42	1.00
BDT no-Bin: ICA-5	1831.83 ± 7.10	1933.91 ± 6.40	25.45 ± 0.42%	51.25 ± 0.18%	4.00e−03	5	2663.76	0.23
BDT no-Bin: PCA-10	1779.06 ± 7.01	1931.01 ± 6.40	27.60 ± 0.41%	51.32 ± 0.18%	6.92e−03	10	2625.62	0.23
BDT no-Bin: Rings[1:100]	1384.80 ± 5.26	1579.93 ± 5.24	43.64 ± 0.31%	60.17 ± 0.15%	5.95e−02	100	2100.92	1.03
RFBDT no-Bin: Rings[1:100]	1390.96 ± 5.76	1562.75 ± 5.19	43.39 ± 0.33%	60.61 ± 0.15%	2.08e−01	100	2092.11	1.29

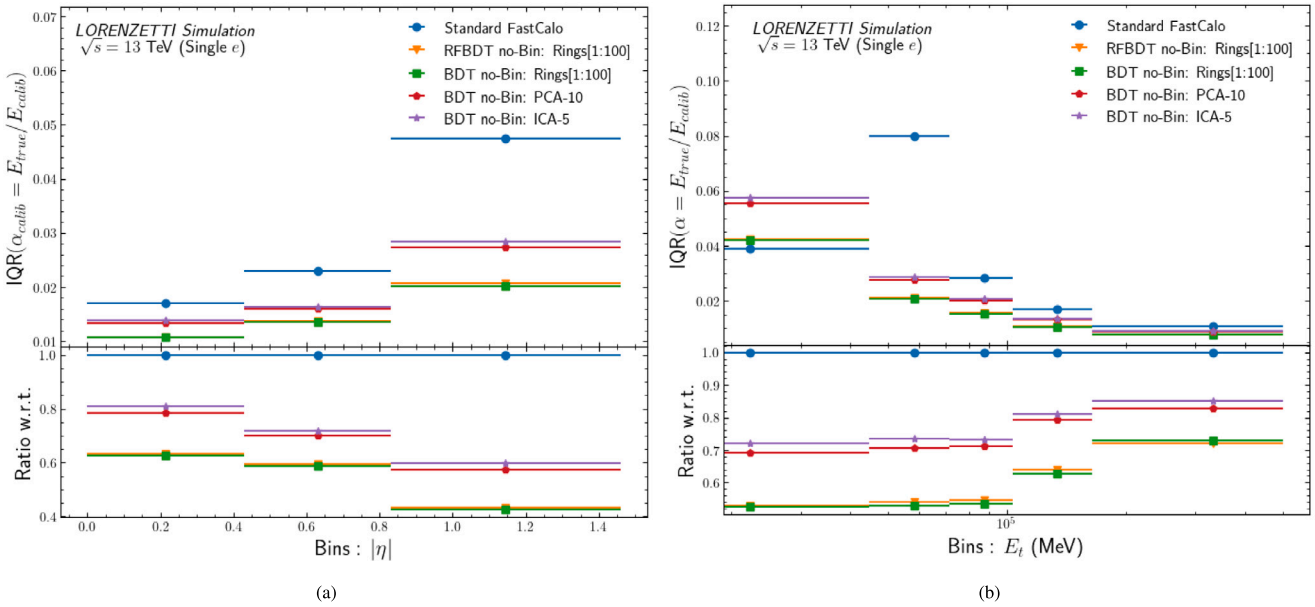


Fig. 12. Bin-wise comparison of the interquartile range of the calibration factor, IQR(α), for the top-ranked calibration methods across (a) $|\eta|$ and (b) transverse-energy E_T bins. The upper panels show the absolute IQR values, while the lower panels show the values relative to the Standard FastCalo reference. In the relative representation, the Standard FastCalo reference is fixed at 1.0, and values below this baseline indicate reduced dispersion of the calibration-factor distribution.

ods generally follow the same global trend as the reference calibration factor, with the largest relative gains observed outside the lowest transverse-energy interval.

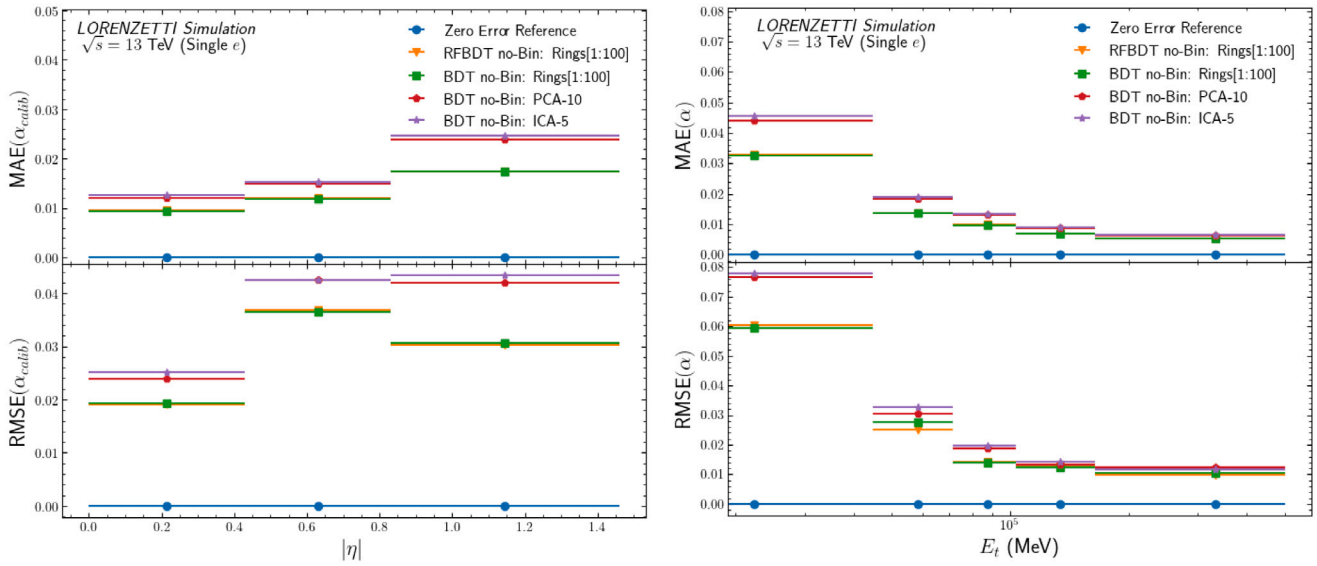
It is important to note that, in the lowest transverse-energy interval, the Standard FastCalo baseline may appear locally more stable than some ML-based configurations in terms of IQR(α). This does not contradict the global calibration results, because this plot evaluates the dispersion of the calibration factor α, rather than the direct calibrated-energy error. At low transverse energies, the calibration-factor ratio defined in Eq. (9) becomes more sensitive to small absolute fluctuations in E_{fast}, since E_{fast} appears in the denominator of the multiplicative correction factor. According to the standard law of propagation of uncertainty for quotient-based measurement models, the relative uncertainty of a ratio is directly affected by the relative uncertainty of its denominator [62]. Therefore, when E_{fast} is small, even modest absolute fluctuations may produce comparatively larger variations in α_{true}. In calorimetric measurements, this effect is further amplified at low energy because the stochastic and noise terms of the energy resolution have a larger relative contribution in this regime [32,33,63]. Consequently, estimating multiplicative correction factors is intrinsically more challenging in the lowest E_T bin, which explains why a limited local advantage of the baseline can appear in IQR(α) without invalidating the overall improvement achieved by the ML-based calibrators.

In addition, the ML models were trained globally to minimize the calibrated-energy error over the full phase space, rather than to optimize IQR(α) independently in each energy bin. Therefore, a limited local advantage of the baseline in the first E_T interval is compatible with the overall improvement achieved by the ML-based calibrators, particularly in the higher-energy regions where nonlinear response effects are more pronounced.

Fig. 13 complements this analysis by reporting the MAE and RMSE of the predicted calibration factors with respect to α_{true}. The same regional behavior is observed: both error metrics increase with $|\eta|$ and decrease with increasing E_T. This indicates that the calibration task becomes more challenging in regions of larger pseudorapidity and lower transverse energy, where detector-response variations and relative fluctuations are more pronounced. Nevertheless, the selected ML configurations maintain consistent behavior across the evaluated bins, suggesting that the globally trained models do not introduce strong localized instabilities.

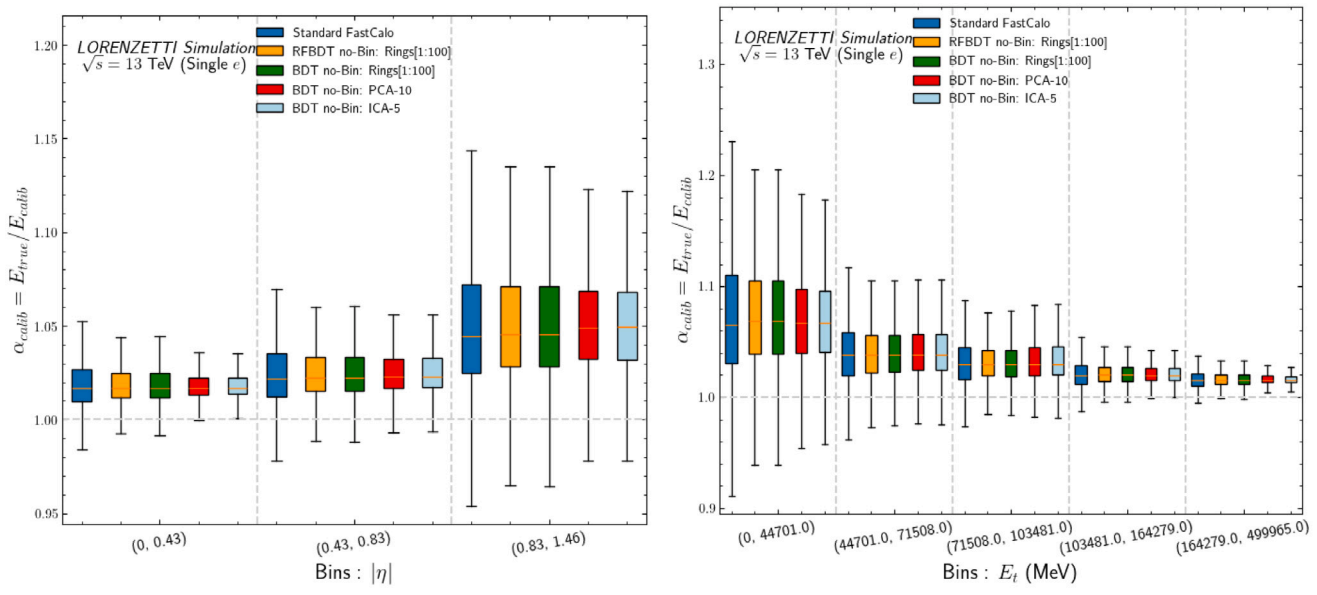
The distributions of the predicted calibration factors are further summarized in Fig. 14. The boxplots confirm the trends observed in the IQR, MAE, and RMSE analyses. The dispersion of α_{calib} increases with $|\eta|$, whereas it decreases with increasing E_T. Importantly, the medians of the ML-predicted calibration factors remain aligned with the reference behavior across the evaluated regions, indicating that the globally trained models preserve the main regional dependence of the calibration correction while reducing residual variability.

These results support the use of global, non-binned training in the present study. Although bin-specific training can improve local specialization, it may also introduce artificial discontinuities at bin boundaries



(a) (b)

Fig. 13. Bin-wise comparison of the MAE and RMSE of the predicted calibration factors with respect to the target correction factor $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$, evaluated across (a) $|\eta|$ and (b) E_T bins. The Zero Error Reference corresponds to the ideal case in which the predicted calibration factor is exactly equal to the target value, i.e., $\alpha_{\text{calib}} = \alpha_{\text{true}}$, and is included only as a visual reference for zero error. Since α_{true} is defined from the Standard FastCalo estimate, the Standard FastCalo response provides the reference target used to compute the calibration-factor errors.



(a) (b)

Fig. 14. Boxplots of the predicted calibration factors α_{calib} for the top-ranked methods across (a) $|\eta|$ and (b) E_T bins. The box width represents the interquartile range, and the dashed vertical lines indicate the bin boundaries.

and reduce statistical robustness in sparsely populated regions. By contrast, the global models learn a continuous calibration function over the full feature space, while the bin-wise evaluation provides an a posteriori diagnostic of regional stability. This distinction is important: the bins are not used to train independent models, but rather to verify whether the globally trained calibrators remain reliable across relevant detector and energy regimes.

A more complete evaluation of segmentation-aware calibration is left for future work. Such an analysis would require larger simulated samples with more homogeneous populations across the $(|\eta|, E_T)$ regions, allowing local models to be trained without excessive statistical fragmentation. This future direction will help determine whether region-adaptive calibration can provide additional gains beyond the global hybrid strategies evaluated here.

3.3. Energy resolution

Energy resolution provides an energy-dependent assessment of the precision of the calibrated response and complements the global RMSE and IQR analyses. While the global metrics summarize the overall

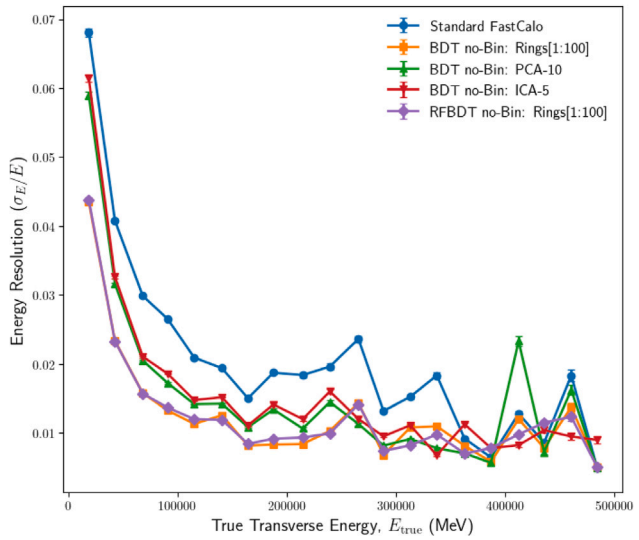


Fig. 15. Relative energy resolution, expressed as σ_E/E, as a function of the true transverse energy E_{true} for the Standard FastCalo baseline and the four top-ranked ML-based calibration methods. The curves show the energy-dependent dispersion of the calibrated response over the full range considered in this study. The ML-based methods provide systematically improved resolution with respect to Standard FastCalo, especially in the low- and intermediate-energy regions.

agreement between E_{calib} and E_{true}, the resolution curve shows how the dispersion of the reconstructed response evolves across the full transverse-energy range. This is particularly relevant for calorimetric measurements, where the relative impact of noise, shower fluctuations, and reconstruction nonlinearities is generally more pronounced at low and intermediate energies.

Fig. 15 shows the relative energy resolution as a function of E_{true} for the Standard FastCalo baseline and the four selected ML-based calibration methods. All ML configurations improve the resolution over most of the analyzed spectrum, demonstrating that the reduction in calibration error observed in the global RMSE and IQR metrics also translates into improved energy-dependent precision. The largest gains are observed in the low- and intermediate-energy regions, where fluctuations and reconstruction biases have a stronger relative impact on the measured response. At higher energies, the differences between methods become less pronounced, but the ML-based calibrators remain competitive with or superior to the baseline.

Among the evaluated methods, RFBDT `no-Bin: Rings[1:100]` exhibits the most favorable overall behavior. This configuration combines the full 100-ring representation with a Random Forest baseline and BDT residual-correction stage, yielding low resolution values and a smooth evolution across the energy range. The BDT `no-Bin: Rings[1:100]` configuration follows closely, confirming that preserving the complete ring topology provides the strongest absolute calibration performance.

The compact configurations, BDT `no-Bin: PCA-10` and BDT `no-Bin: ICA-5`, also substantially improve the resolution relative to Standard FastCalo despite using only 10 and 5 input components, respectively. Their performance confirms that low-dimensional latent representations retain much of the calibration-relevant information contained in the full ring profile. Some larger local fluctuations are observed at higher energies, particularly for BDT `no-Bin: PCA-10`, which is consistent with the reduced input dimensionality and the smaller number of events typically available in the upper energy range.

Overall, the energy-resolution analysis confirms that the proposed ML-based calibrators improve not only the average reconstruction accuracy but also the precision of the calorimetric response as a function of energy. From an operational perspective, this improvement is important because a narrower response distribution reduces event migration across selection thresholds and contributes to more stable trigger-level energy reconstruction.

3.4. Turn-On curves

Turn-on curves are used to assess how improvements in energy calibration affect threshold-based event selection. While RMSE, IQR, and energy resolution quantify the accuracy and dispersion of the reconstructed energy, the turn-on analysis evaluates the operational impact of the calibration in the near-threshold region, where trigger decisions are most sensitive to reconstruction errors.

Fig. 16 shows the event acceptance rate (EAR) as a function of the threshold offset $\Delta\lambda$ for the Standard FastCalo baseline and two representative ML-based calibrators. In this representation, a smaller value of $\Delta\lambda$ required to reach a given acceptance level indicates a sharper turn-on curve and, consequently, a more efficient threshold response. This behavior is particularly important for trigger-level applications, since a broad transition region increases the probability of event migration across the selection threshold.

A clear improvement over the baseline is observed for both ML-based configurations. The full-ring hybrid model, RFBDT `no-Bin: Rings[1:100]`, reaches 90% event acceptance at $\Delta\lambda = 1858.80$ MeV, whereas Standard FastCalo requires $\Delta\lambda = 5491.13$ MeV to reach the same operating point. This corresponds to a reduction of 66.15% in $\Delta\lambda_{90}$. At a fixed offset of $\Delta\lambda = 1637.11$ MeV, the same ML-based method achieves an EAR of 86%, compared with only 29% for the baseline.

The compact configuration BDT `no-Bin: ICA-5` also produces a substantially sharper turn-on response. It reaches 90% acceptance at $\Delta\lambda = 2421.55$ MeV, corresponding to a 55.90% reduction relative to Standard FastCalo. At $\Delta\lambda = 1807.64$ MeV, this compact model achieves an EAR of 80%, while the baseline reaches only 32%. These results show that even a five-component latent representation retains enough calibration information to improve threshold behavior substantially.

The full set of energy-resolution and turn-on metrics is summarized in Table 5. The two full-ring configurations, BDT `no-Bin: Rings[1:100]` and RFBDT `no-Bin: Rings[1:100]`, achieve the strongest turn-on improvement, both reducing $\Delta\lambda_{90}$ by 66.15%. The compact configurations, BDT `no-Bin: PCA-10` and BDT `no-Bin: ICA-5`, remain competitive, reducing $\Delta\lambda_{90}$ by 57.14% and 55.90%, respectively. This ordering is consistent with the energy-resolution analysis: the full-ring models provide the lowest σ_E/E, while the compact PCA- and ICA-based models preserve much of the gain with substantially fewer input features.

Overall, the ML-based calibrators produce sharper near-threshold selection curves than Standard FastCalo. From an operational perspective, the reduction in $\Delta\lambda_{90}$ indicates a narrower transition region between event rejection and acceptance, reducing event migration across trigger thresholds. These results demonstrate that the proposed calibration strategies improve not only the precision of reconstructed energy measurements, but also the practical efficiency of threshold-based event selection.

3.5. Integrated performance and accuracy–efficiency analysis

While the previous analyses quantified the improvements in terms of global error metrics, energy resolution, turn-on response, statistical significance, and computational efficiency, the figures discussed here summarize these results from complementary perspectives. First, scatter plots are used to inspect the event-by-event linearity and dispersion of the calibrated energy response. Then, a compact performance summary compares the Standard FastCalo baseline with the best full-ring and compact configurations across the main calibration metrics. Finally, an accuracy–efficiency trade-off plot relates calibration quality to trained-model prediction time and input dimensionality, highlighting

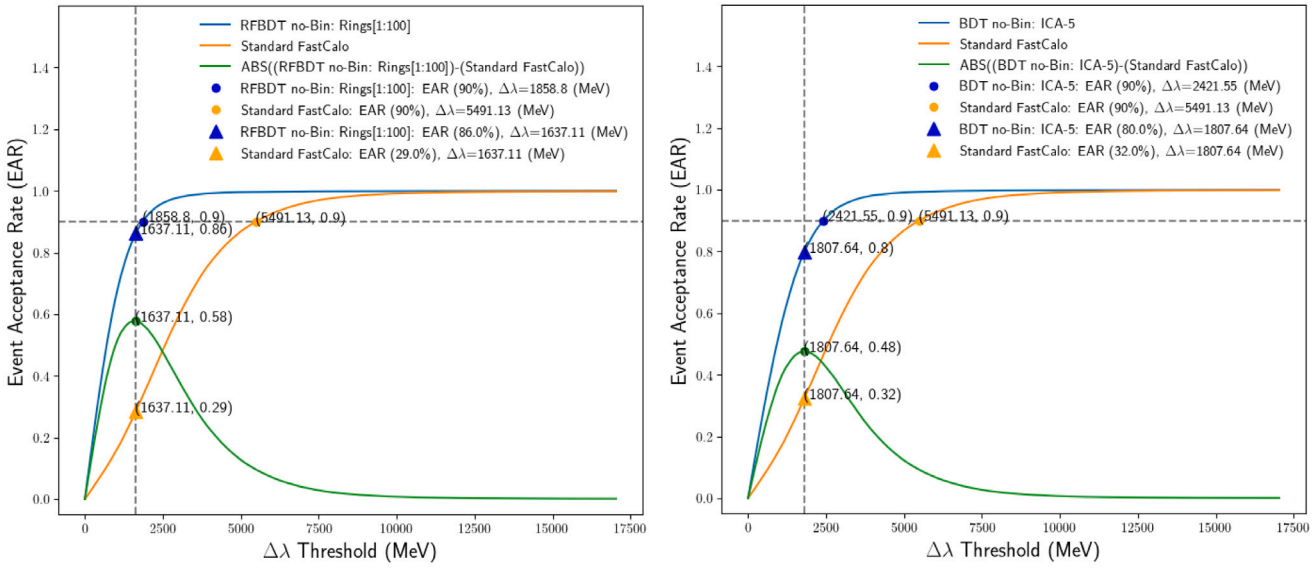


Fig. 16. Trigger turn-on curves showing the event acceptance rate (EAR) as a function of the threshold offset $\Delta\lambda$ for Standard FastCalo and two representative ML-based calibration methods. Panel (a) compares RFBDT `no-Bin: Rings[1:100]` with Standard FastCalo, while panel (b) compares BDT `no-Bin: ICA-5` with Standard FastCalo. The green curve represents the absolute difference in EAR between the ML-based method and the baseline. The dashed lines and annotated markers indicate representative operating points used to quantify the threshold offset required to reach a given acceptance level and the gain in acceptance at fixed $\Delta\lambda$.

Table 5
Energy-resolution and turn-on-curve summary for Standard FastCalo and the four highest-ranked ML-based calibration configurations. The reported quantities are the relative energy resolution σ_E/E, its relative improvement with respect to Standard FastCalo, the threshold offset $\Delta\lambda_{90}$ required to achieve 90% event acceptance, and the corresponding relative reduction in $\Delta\lambda_{90}$ with respect to the baseline. All results were obtained from the independent test set, corresponding to 30% of the full dataset. The values of σ_E/E are reported as mean ± standard deviation, and Standard FastCalo is used as the reference baseline for all percentage comparisons.

Method	$\frac{\sigma_E}{E}$	$\frac{\sigma_E}{E}$ (% Std)	$\Delta\lambda_{90}$ (MeV)	$\Delta\lambda_{90}$ (% Std)
Standard FastCalo	2.46e−02 ± 5.80e−05	–	5491.13	–
BDT no-Bin: ICA-5	1.67e−02 ± 3.90e−05	31.98% ± 0.23%	2421.55	55.90%
BDT no-Bin: PCA-10	1.67e−02 ± 3.90e−05	32.05% ± 0.23%	2353.34	57.14%
BDT no-Bin: Rings[1:100]	1.37e−02 ± 3.20e−05	44.36% ± 0.18%	1858.80	66.15%
RFBDT no-Bin: Rings[1:100]	1.35e−02 ± 3.20e−05	44.96% ± 0.18%	1858.80	66.15%

the benchmark-level compromise between predictive performance and CPU-based inference cost.

Fig. 17 compares Standard FastCalo with two representative ML-based calibration methods. In both panels, the dashed diagonal line corresponds to the ideal response, $E_{\text{calib}} = E_{\text{true}}$. The Standard FastCalo distribution exhibits a broader spread around this line, reflecting the nonlinearities and event-by-event fluctuations already observed in the uncalibrated response. In contrast, the ML-based predictions are more tightly concentrated around the diagonal, indicating improved agreement with the reference energy and reduced dispersion of the calibrated response. The full-ring configuration, RFBDT `no-Bin: Rings[1:100]`, shows the most favorable visual behavior. Its point cloud is more concentrated around the ideal response and exhibits fewer strongly miscalibrated events relative to Standard FastCalo. This observation is consistent with its leading performance in the global ranking, energy-resolution analysis, and turn-on-curve evaluation. The compact configuration, BDT `no-Bin: ICA-5`, also provides a clear improvement over the baseline despite relying on only five input components. Although its scatter distribution remains slightly broader than that obtained with the full-ring model, it still follows the ideal response more closely than Standard FastCalo across the analyzed energy range. This confirms that the low-dimensional ICA representation retains a substantial fraction of the calibration-relevant information contained in the full ring profile. The scatter-plot analysis reinforces the conclusions obtained from the quantitative metrics: the proposed ML-based calibrators reduce the spread of reconstructed energies around the ideal response, improve the linearity of the calibrated measurement, and decrease the occurrence of large event-by-event deviations.

To provide a more direct visual summary of the main calibration gains, Fig. 18 compares representative top–performing configurations across four complementary performance indicators: RMSE, IQR, relative energy resolution σ_E/E, and the turn-on threshold offset $\Delta\lambda_{90}$. The values are expressed as percentage improvements relative to the Standard FastCalo baseline; therefore, higher values indicate greater improvement. This representation shows that the full-ring configurations, particularly RFBDT `no-Bin: Rings[1:100]` and BDT `no-Bin: Rings[1:100]`, provide the largest overall gains, especially in RMSE and turn-on response. Compact latent-space configurations, such as BDT `no-Bin: ICA-5` and BDT `no-Bin: PCA-10`, preserve competitive improvements while relying on substantially lower-dimensional input representations.

A complementary view of the accuracy–efficiency trade-off is shown in Fig. 19. The vertical axis represents the joint RMSE–IQR metric d_0, for which lower values indicate better calibration accuracy and lower error dispersion, whereas the horizontal axis shows the mean model prediction time per event in logarithmic scale. The marker

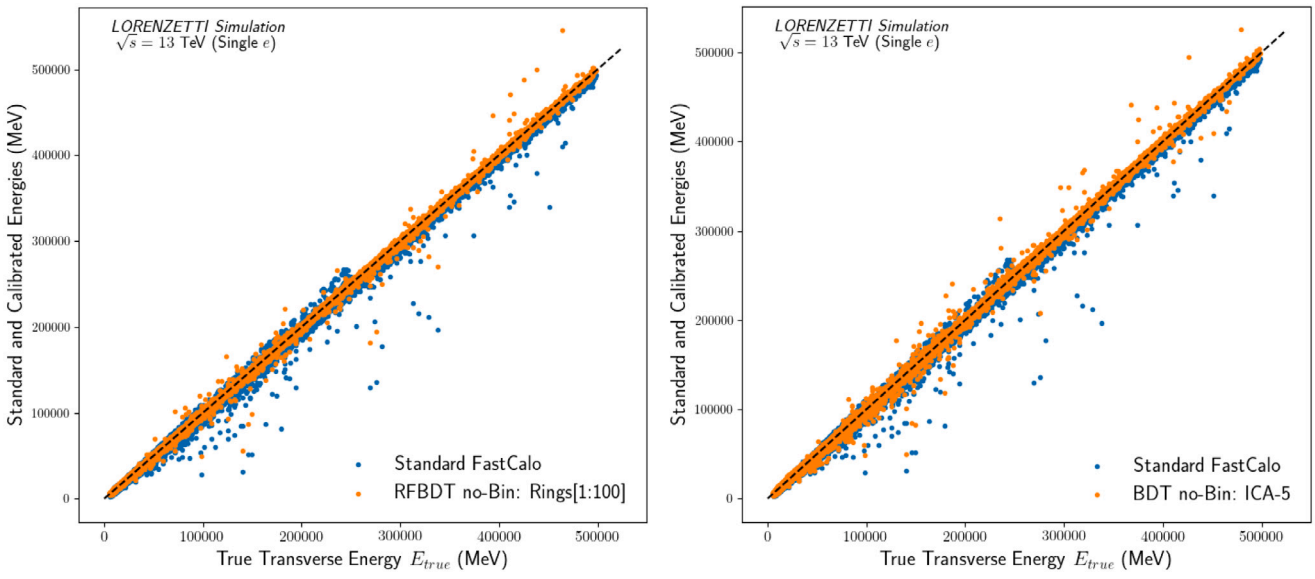


Fig. 17. Scatter plots of calibrated transverse energy as a function of true transverse energy for Standard FastCalo and two representative ML-based calibration methods. Panel (a) compares Standard FastCalo with RFBDT `no-Bin: Rings[1:100]`, while panel (b) compares Standard FastCalo with BDT `no-Bin: ICA-5`. The dashed diagonal line indicates the ideal response, $E_{\text{calib}} = E_{\text{true}}$. In both panels, the ML-based calibrators show a tighter concentration around the diagonal and a reduced spread of large deviations relative to Standard FastCalo, indicating improved linearity, accuracy, and event-by-event precision over the full energy range.

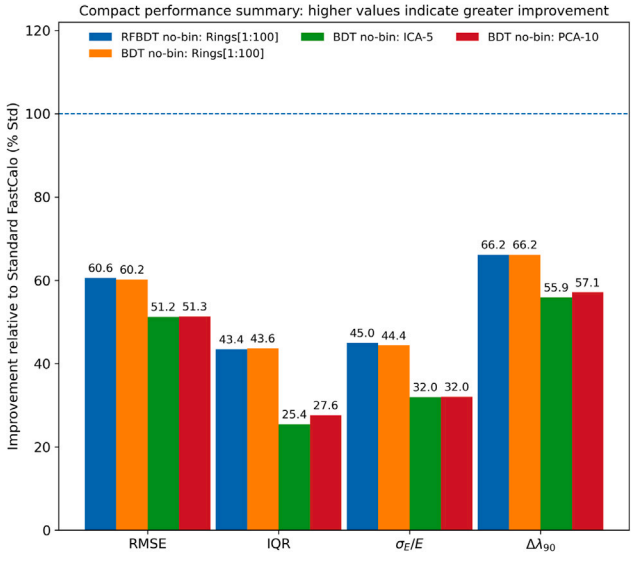


Fig. 18. Compact performance summary of representative top-performing calibration configurations. Bars show the percentage improvement relative to the Standard FastCalo baseline for RMSE, IQR, relative energy resolution σ_E/E, and the turn-on threshold offset $\Delta\lambda_{90}$. Higher values indicate greater improvement. The full-ring configurations, RFBDT `no-Bin: Rings[1:100]` and BDT `no-Bin: Rings[1:100]`, achieve the largest overall gains, while the compact latent-space models, BDT `no-Bin: ICA-5` and BDT `no-Bin: PCA-10`, retain competitive performance with reduced input dimensionality.

size encodes the number of input features used by each configuration, and the dashed horizontal line indicates the Standard FastCalo d_0 reference. This representation makes explicit the benchmark-level compromise between calibration accuracy, trained-model prediction time, and input dimensionality. Compact latent-space models such as BDT `no-Bin: ICA-5` and BDT `no-Bin: PCA-10` require only 5 and 10 input features, respectively, and operate at very low inference times, although with higher d_0 values. In contrast, the full-ring models use 100 input features and achieve substantially lower d_0 values, remaining well below the Standard FastCalo reference. Among them, RFBDT `no-Bin: Rings[1:100]` provides the lowest d_0, while BDT `no-Bin: Rings[1:100]` offers a similar accuracy level with lower inference time.

The integrated performance analysis confirms that the proposed machine-learning calibrators improve the simulated fast calorimetric energy measurement from both accuracy-oriented and CPU-based efficiency perspectives. The full-ring RFBDT configuration provides the strongest calibration gains and the lowest joint error metric, whereas compact latent-space models such as BDT `ICA-5` preserve competitive performance with substantially reduced input dimensionality and very low inference time. These results reinforce the central benchmark-level conclusion of this work: hybrid tree-based calibration can substantially improve simulated fast particle-detector energy measurements while preserving computational characteristics that motivate further studies in dedicated low-latency reconstruction environments.

3.6. Statistical validation

The Normality (Gaussianity) assessment for $\epsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$ and $\epsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$, as well as for their paired differences $d = \epsilon_{\text{calib}} - \epsilon_{\text{fast}}$, using the D'Agostino–Pearson and Anderson–Darling tests, yielded p-values $< 10^{-16}$ in all configurations. This led to rejection of the null hypothesis of normality for every distribution examined. Representative Q–Q plots for the RFBDT `no-Bin: Rings[1:100]` and BDT `no-Bin: ICA-05` configuration (Fig. 20) show marked deviations from the $y = x$ line in both tails, consistent with heavy-tailed and asymmetric behavior, a pattern also observed for the other top-ranked methods.

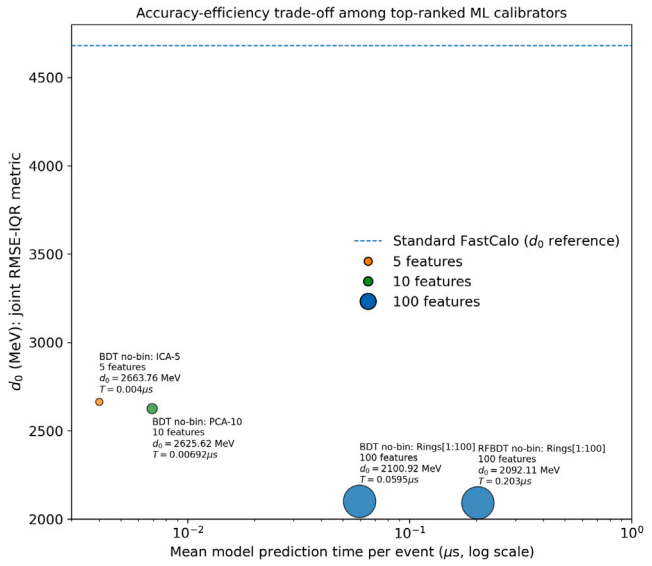


Fig. 19. Accuracy–efficiency trade-off among representative top-ranked machine-learning calibrators. The vertical axis shows the joint RMSE–IQR metric d_0, where lower values indicate better calibration accuracy and error dispersion. The horizontal axis shows the mean model prediction time per event in logarithmic scale. Marker size represents the number of input features used by each configuration, and the dashed horizontal line indicates the Standard FastCalo d_0 reference. Compact latent-space models achieve the lowest CPU-based prediction times with 5 or 10 input features, whereas full-ring models achieve the lowest d_0 values using 100 input features.

Although the residual distributions are non-Gaussian, the calibrated and uncalibrated errors are strictly paired because they are computed from the same events. Therefore, paired statistical comparisons are appropriate, with non-parametric and bootstrap-based tests used to ensure robustness against deviations from normality, as described in the Methodology section.

Non-parametric methods were also applied to confirm robustness against distributional departures, including the Wilcoxon signed-rank test for central tendency and the paired bootstrap BCa test for variance ratios.

3.6.1. Mean error comparison

Table 6 summarizes the results of three complementary approaches applied to the paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$: the large-sample paired z-test (primary), the paired Student's t-test (reference), and the Wilcoxon signed-rank test (robustness check).

Across all configurations and test types, the null hypothesis $H_0 : \mu_d \geq 0$ was rejected at $\alpha = 0.05$ with overwhelming statistical significance. For both the z- and t-tests, the p-values were below the numerical precision of the machine, reported as $p < 10^{-300}$. The Wilcoxon signed-rank test, which does not require normality, yielded consistent rejections. In all cases, the one-sided 95% confidence intervals for μ_d (or for the median of d in the Wilcoxon test) lay entirely below zero, which is the interval-based equivalent of rejecting H_0.

From a physical standpoint, negative values of μ_d indicate that calibration systematically reduces the residuals compared to the uncalibrated method. For RFBDT `no-Bin: Rings[1:100]`, $\mu_d = -2.76$ GeV (Table 6), meaning that, on average, the residual after calibration is at least 2.76 GeV closer to zero for the same event. Comparable magnitudes of improvement in μ_d are observed for BDT `no-Bin: Rings[1:100]`, BDT `no-Bin: PCA-10`, and BDT `no-Bin: ICA-5`, confirming that the effect is consistent across all ranked configurations.

3.6.2. Variance ratio analysis

Table 7 presents the results of three complementary approaches for testing the variance ratio $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$: the Pitman–Morgan test (primary), the large-sample z approximation F-test (comparative reference), and the paired bootstrap BCa test (robustness check). The F-test is included solely for comparability with traditional independent-sample methods, despite not accounting for the paired nature of the data.

Across all configurations, the null hypothesis $H_0 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} \geq 1$ was rejected with overwhelming statistical significance. For the Pitman-Morgan and F-tests, p-values fell below the numerical precision of the machine and are reported as $p < 10^{-300}$. For the paired bootstrap BCa test, the smallest possible p-value is limited by the number of resamples (B), with $p_{\min} = 1/B$; here $B = 5000$ yields $p_{\min} = 0.0002$, which explains why the reported value is not as extreme as in the parametric tests.

Negative PM-statistics occur because the variance of the calibrated errors is consistently smaller than that of the uncalibrated errors. This sign change arises from the fact that the Pitman–Morgan statistic is proportional to the correlation between the sums and differences of paired observations, which becomes negative when the first variance is lower than the second.

The *paired bootstrap BCa* approach builds confidence intervals without assuming normality, making it robust to heavy-tailed or skewed error distributions. Its "paired" design preserves the intrinsic correlation between $\varepsilon_{\text{calib}}$ and $\varepsilon_{\text{fast}}$ by resampling event pairs jointly. The intervals obtained matched the conclusions from the parametric tests, reinforcing the robustness of the variance reduction findings.

It is important to note that the variance ratio statistic shown for the bootstrap test (e.g., 0.3029) is identical to the F-statistic. This is not an inconsistency but rather a fundamental feature of the bootstrap methodology. The bootstrap procedure uses the original sample's calculated statistic (in this case, $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$) as its starting point. The difference between the tests lies not in this initial value, but in the method of inference: the F-test derives its p-value from a theoretical probability distribution, while the bootstrap derives it empirically by resampling the observed data. The convergence of both methods towards the same conclusion, despite their radically different assumptions, underscores the robustness and veracity of the finding.

For RFBDT `no-Bin: Rings[1:100]`, the variance after calibration is approximately 33% of the uncalibrated variance, corresponding to a 67% reduction. Similar reductions are observed for BDT `no-Bin: Rings[1:100]` (66%), BDT `no-Bin: PCA-10` (47%), and BDT `no-Bin: ICA-5` (49%), confirming that the reduction in variance is consistent across all BDT configurations.

3.6.3. Statistical interpretation

The joint occurrence of negative μ_d and variance ratios well below unity indicates that the calibration achieves both a systematic bias reduction (mean residuals closer to zero) and a variance reduction (tighter residual dispersion) relative to the uncalibrated baseline. From a statistical standpoint, these two effects act synergistically to reduce the mean squared error (MSE) of the reconstructed energy [22,46]:

$$\text{MSE} = \text{Var}(\varepsilon) + [\text{Bias}(\varepsilon)]^2.$$

Since both $\text{Var}(\varepsilon)$ and $|\text{Bias}(\varepsilon)|$ are smaller after calibration, the overall MSE is necessarily reduced. This implies that the reconstructed energy distributions become both narrower and more accurately centered around the true value.

From a physics perspective, such simultaneous improvements in bias and resolution enhance the calorimeter's resolving power, benefiting tasks such as resonance peak identification, suppression of background fluctuations, and precision cross-section measurements, where small biases and high resolution are critical to achieving optimal sensitivity.

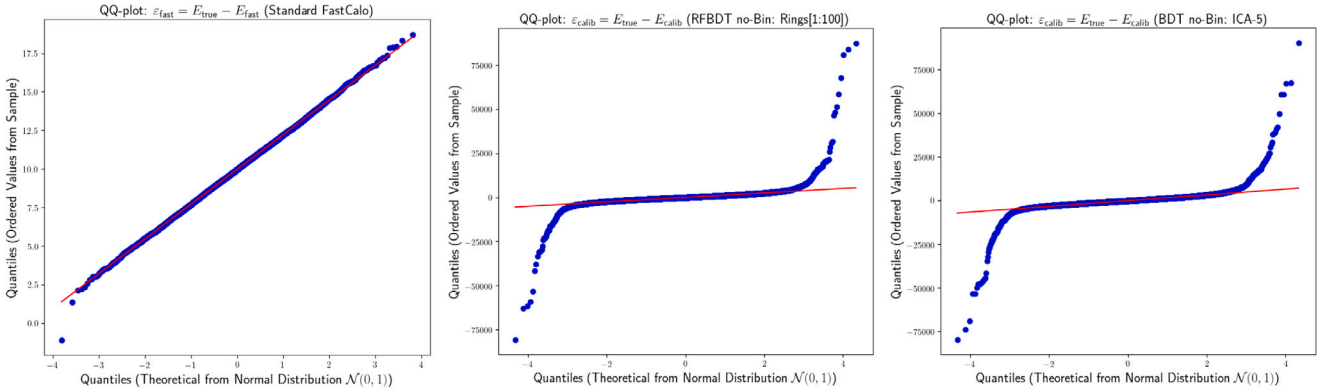


Fig. 20. Representative Q–Q plots for $\varepsilon_{\text{fast}}$ and $\varepsilon_{\text{calib}}$ with respect to a standard normal distribution. Panel (a) shows the Standard FastCalo residuals, while panels (b) and (c) show the calibrated residuals for RFBDT `no-Bin: Rings[1:100]` and BDT `no-Bin: ICA-5`, respectively. In each panel, the blue points represent the ordered empirical quantiles of the corresponding residual distribution, and the red line denotes the theoretical reference line expected under normality. Deviations of the blue points from the red line, especially in the distribution tails, indicate departures from Gaussian behavior, including heavy-tailed and asymmetric residual patterns. This departure from normality motivates the use of complementary non-parametric and robust paired statistical methods, such as the Wilcoxon signed-rank test for central tendency and bootstrap-based inference for variance ratios, in addition to classical parametric approaches.

Table 6
Statistical evaluation of mean calibration error using paired one-sided *t*, *z*, and Wilcoxon signed-rank tests. Hypotheses: $H_0 : \mu_d \geq 0$ vs. $H_1 : \mu_d < 0$, where μ_d is the mean (or median for Wilcoxon) of the paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$, with $\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$ and $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$. All confidence intervals are one-sided at 95%. Units for μ_d and CIs are in MeV.

Method (z-test: primary)	*z*-statistic	*p*-value	95% one-sided CI for μ_d [MeV]	Decision
BDT no-Bin: Rings[1:100]	−341.67	$< 10^{-300}$	(-inf, −2759.49)	Reject H0
BDT no-Bin: PCA-10	−303.83	$< 10^{-300}$	(-inf, −2837.66)	Reject H0
BDT no-Bin: ICA-5	−345.97	$< 10^{-300}$	(-inf, −2855.17)	Reject H0
RFBDT no-Bin: Rings[1:100]	−328.64	$< 10^{-300}$	(-inf, −2761.62)	Reject H0
Method (t-test: reference)	*t*-statistic	*p*-value	95% one-sided CI for μ_d [MeV]	Decision
BDT no-Bin: Rings[1:100]	−341.67	$< 10^{-300}$	(-inf, −2759.48)	Reject H0
BDT no-Bin: PCA-10	−303.83	$< 10^{-300}$	(-inf, −2837.66)	Reject H0
BDT no-Bin: ICA-5	−345.97	$< 10^{-300}$	(-inf, −2855.17)	Reject H0
RFBDT no-Bin: Rings[1:100]	−328.64	$< 10^{-300}$	(-inf, −2761.62)	Reject H0
Method (Wilcoxon signed-rank test: robustness)	*W*-statistic	*p*-value	95% one-sided CI for median(d) [MeV]	Decision
BDT no-Bin: Rings[1:100]	79 216 155	$< 10^{-300}$	(-inf, −2546.05)	Reject H0
BDT no-Bin: PCA-10	71 666 715	$< 10^{-300}$	(-inf, −2598.35)	Reject H0
BDT no-Bin: ICA-5	69 189 466	$< 10^{-300}$	(-inf, −2606.25)	Reject H0
RFBDT no-Bin: Rings[1:100]	79 115 535	$< 10^{-300}$	(-inf, −2553.72)	Reject H0

Table 7
Statistical evaluation of the error variance ratio using Pitman–Morgan, one-sided *F*, and paired bootstrap tests. Hypotheses: $H_0 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} \geq 1$ vs. $H_1 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} < 1$, with σ^2_{calib} denoting the variance of $\varepsilon_{\text{calib}}$ and σ^2_{fast} the variance of $\varepsilon_{\text{fast}}$. All confidence intervals are one-sided at 95% and expressed in dimensionless units (variance ratios). Lower bounds of CIs were truncated at 0 to satisfy the non-negativity of variance.

Method (Pitman-Morgan: primary)	PM-statistic	*p*-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
BDT no-Bin: Rings[1:100]	−217.35	$< 10^{-300}$	(0.0, 0.3416)	Reject H0
BDT no-Bin: PCA-10	−127.16	$< 10^{-300}$	(0.0, 0.5319)	Reject H0
BDT no-Bin: ICA-5	−137.96	$< 10^{-300}$	(0.0, 0.5122)	Reject H0
RFBDT no-Bin: Rings[1:100]	−214.14	$< 10^{-300}$	(0.0, 0.3325)	Reject H0
Method (F-test: reference)	*F*-statistic	*p*-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
BDT no-Bin: Rings[1:100]	0.3096	$< 10^{-300}$	(0.0, 0.3130)	Reject H0
BDT no-Bin: PCA-10	0.4617	$< 10^{-300}$	(0.0, 0.4667)	Reject H0
BDT no-Bin: ICA-5	0.4627	$< 10^{-300}$	(0.0, 0.4677)	Reject H0
RFBDT no-Bin: Rings[1:100]	0.3029	$< 10^{-300}$	(0.0, 0.3062)	Reject H0
Method (Paired bootstrap BCa test: robustness)	$(\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}})$-statistic	*p*-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
BDT no-Bin: Rings[1:100]	0.3096	0.0002	(0.0, 0.3405)	Reject H0
BDT no-Bin: PCA-10	0.4617	0.0002	(0.0, 0.5315)	Reject H0
BDT no-Bin: ICA-5	0.4627	0.0002	(0.0, 0.5120)	Reject H0
RFBDT no-Bin: Rings[1:100]	0.3029	0.0002	(0.0, 0.3327)	Reject H0

3.7. Impact of pile-Up on calibration performance

Pile-up is a major source of degradation in calorimetric energy measurements. Additional energy deposits unrelated to the primary hard-scatter process may bias the reconstructed transverse energy, broaden the response distribution, distort the ring-energy topology, and reduce the stability of threshold-based event selection. These effects become increasingly relevant in high-luminosity conditions, where the

average number of interactions per bunch crossing, $\langle\mu\rangle$, can reach several tens.

As described in Section 2.1, the Lorenzetti simulation used in this work includes pile-up contributions by merging minimum-bias events into the primary event. The simulated pile-up distribution shown in Fig. 1 has mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999, indicating that the evaluated calibration models were trained and tested under a pile-up-inclusive benchmark with substantial additional collision activity. This level is comparable to high-pile-up Run 2 conditions: ATLAS has reported an average of about 34 independent inelastic proton–proton collisions per bunch crossing for the 2015–2018 data-taking period, while other Run 2 summaries report yearly average pile-up values reaching the upper 30s. The upper tail of the Lorenzetti distribution also extends toward $\langle\mu\rangle \sim 60$–70, approaching the scale of Run 3 sustained pile-up conditions, for which ATLAS reports operation with an average of up to about 60 interactions per bunch crossing [43,64]. Therefore, the reported RMSE, IQR, energy-resolution, and turn-on results already include the pile-up treatment implemented in the adopted Lorenzetti configuration.

However, the present study does not perform a dedicated scan of calibration performance as a function of $\langle\mu\rangle$. Consequently, the reported results should be interpreted as an assessment obtained under the simulated pile-up distribution of the Lorenzetti benchmark, rather than as an explicit pile-up-dependence study. In particular, the compact PCA- and ICA-based representations should not be interpreted as intrinsically suppressing pile-up noise or guaranteeing invariance to different pile-up levels. Their role in this work is to reduce the dimensionality of the ring-based input space while preserving calibration-relevant information.

A dedicated study varying the pile-up level would be required to quantify this dependence explicitly. Such an analysis should evaluate RMSE, IQR, energy resolution, and trigger turn-on behavior in samples stratified by controlled $\langle\mu\rangle$ regimes, including low, medium, and high-pile-up conditions.

4. Conclusions

This work presented a systematic simulation-based evaluation of machine-learning-based strategies for fast transverse-energy calibration in high-energy physics calorimetry. Building on the Lorenzetti-based calibration framework previously established in [21], the present study extended the analysis from conventional regressors and binning strategies to hybrid tree-based models, fixed-cardinality feature representations, and a broader performance-evaluation protocol. The proposed approach was assessed within the considered Lorenzetti simulation benchmark using ring-based shower representations derived from the Ringer algorithm.

The main methodological extension was the evaluation of Random Forest regression and three hybrid tree-based calibration architectures: BDTRF, in which a BDT is followed by Random Forest residual correction; RFBDT, in which a Random Forest baseline is refined by a BDT residual-correction stage; and RFBDTS, a stacked Random Forest–BDT ensemble with a linear meta-learner. In parallel, feature selection and extraction were investigated through correlation ranking, mutual-information ranking, PCA, and ICA, using fixed-cardinality configurations to enable controlled comparisons between predictive performance, model compactness, and inference cost.

Among all evaluated configurations, the best absolute performance was achieved by RFBDT `no-Bin: Rings[1:100]`, which combines a Random Forest baseline predictor with a BDT residual-correction stage while preserving the full 100-ring representation. Relative to the Standard FastCalo baseline under the simulated detector-response conditions considered here, this configuration achieved average reductions of 60.61% in RMSE, 43.39% in IQR, 44.96% in relative energy resolution, and 66.15% in the threshold offset required to reach 90% event acceptance, $\Delta\lambda_{90}$. These results indicate that the proposed residual-learning strategy improves not only the average calibration accuracy, but also the dispersion of the reconstructed response and the sharpness of near-threshold event selection.

From the perspective of compact benchmark-level calibration, BDT `no-Bin: ICA-5` provided the most favorable low-dimensional configuration. Despite using only five input components, it achieved average reductions of 51.25% in RMSE, 25.45% in IQR, 31.98% in relative energy resolution, and 55.90% in $\Delta\lambda_{90}$ relative to Standard FastCalo. This result shows that a substantial fraction of the calibration gain can be retained under aggressive dimensionality reduction, supporting the investigation of compact latent-space representations when input dimensionality and CPU-based prediction cost are relevant design constraints.

The results also reveal a clear trade-off between absolute accuracy and compactness. Full-ring configurations, especially RFBDT `no-Bin: Rings[1:100]` and BDT `no-Bin: Rings[1:100]`, provided the strongest calibration performance. In contrast, compact configurations such as BDT `no-Bin: ICA-5` and BDT `no-Bin: PCA-10` offered more favorable accuracy–efficiency compromises when feature count and CPU-based prediction time were considered together. This behavior was consistently reflected by the global ranking metrics, energy-resolution curves, scatter-plot analyses, and turn-on results.

All four top-ranked configurations, namely RFBDT `no-Bin: Rings[1:100]`, BDT `no-Bin: Rings[1:100]`, BDT `no-Bin: ICA-5`, and BDT `no-Bin: PCA-10`, exhibited mean trained-model prediction times below 7 μs per event on conventional CPU-based hardware. Although this benchmark does not by itself establish readiness for direct deployment in a production trigger chain, it indicates that the evaluated models are computationally lightweight in the relative CPU-based comparison and motivates further latency studies in dedicated online-processing environments.

The statistical validation further supports the consistency of the observed gains within the considered benchmark. Improvements were consistently observed across RMSE, IQR, energy resolution, and threshold-response indicators, showing that the proposed calibrators reduce the spread of residual distributions, improve agreement with the reference energy, and narrow the transition region between event rejection and acceptance. From a measurement perspective, this corresponds to a more stable calibrated response and reduced event migration across the evaluated threshold conditions.

An additional outcome of this work is the demonstration that the Lorenzetti Showers framework provides a flexible and reproducible environment for developing and comparing ML-based calorimeter calibration strategies. Although an ATLAS-inspired electromagnetic calorimeter was used as the reference configuration, similar calibration pipelines could in principle be evaluated for other detector layouts and readout schemes that can be represented within Lorenzetti. However, the present results do not establish detector-independent generalizability; such applicability would require dedicated detector-specific simulations and validation.

The results reported in this work should be interpreted as benchmark-level evidence obtained within the controlled Lorenzetti simulation environment. They demonstrate that ring-based tree and hybrid tree calibration can provide substantial relative improvements over the Standard FastCalo baseline under the simulated detector-response conditions considered here. However, they do not establish readiness for direct deployment in a production high-energy-physics trigger system. Deployment-oriented validation would require evaluation with higher-fidelity detector simulation, collaboration-grade reconstruction chains, experimental data when available, and target-hardware studies that include the complete online feature-construction and inference pipeline, together with realistic memory, transfer, quantization, and worst-case latency constraints.

An important next step is the validation of the proposed calibration methodology using higher-fidelity detector simulations, collaboration-grade reconstruction chains, experimental data when available, and

additional detector-response scenarios. Future studies should also include dedicated simulations with controlled pile-up regimes to quantify the dependence of calibration accuracy, energy resolution, and threshold-response behavior on $\langle\mu\rangle$. Deployment-oriented validation should evaluate the complete feature-construction chain, memory and data-transfer overheads, quantized model implementations, hardware-specific constraints, and worst-case end-to-end latency in trigger-dedicated environments. In addition, the present comparison may be extended to other machine-learning families, including compact neural networks and graph-based models, under the same leakage-free training protocol and measurement-oriented evaluation criteria. Further diagnostics, such as upper-tail residual percentiles, worst-bin summaries, and maximum regional bias, may provide a more detailed characterization of rare-event and localized detector-response behavior.

Within the Lorenzetti benchmark, the results indicate that hybrid tree-based calibration is a promising direction for improving fast calorimetric energy estimation while preserving a favorable balance between predictive performance, model compactness, and relative inference cost.

CRediT authorship contribution statement

Paulo Roberto Araujo da Silva: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Eduardo Furtado de Simas Filho:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation. **Paulo C.M.A. Farias:** Writing – review & editing. **Edmar E.P. Souza:** Writing – review & editing. **Juan L. Marin:** Data curation. **José Manoel de Seixas:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eduardo Furtado de Simas Filho reports article publishing charges and travel were provided by Brazilian National Council for Scientific and Technological Development (Conselho Nacional de Desenvolvimento Científico e Tecnológico - CNPq). Eduardo Furtado de Simas Filho reports article publishing charges and travel were provided by Coordination for the Improvement of Higher Education Personnel (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - CAPES). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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- Brazilian National Council for Scientific and Technological Development (*Conselho Nacional de Desenvolvimento Científico e Tecnológico* - ***CNPq***);
- National Network for High Energy Physics (*Rede Nacional de Física de Altas Energias* - ***RENAFAE***);
- Coordination for the Improvement of Higher Education Personnel (*Coordenação de Aperfeiçoamento de Pessoal de Nível Superior* - ***CAPES***) and French Committee for the Evaluation of University and Scientific Cooperation with Brazil (*Comitê Francês de Avaliação da Cooperação Universitária e Científica com o Brasil* - ***COFECUB***) program.

Appendix A. Lorenzetti repositories

The source code, installation instructions, simulation configurations, and datasets associated with this study are publicly available in the Lorenzetti framework repository [29] and in the dataset repository used in this work [31].

Appendix B. Hyperparameter selection

Tables 8 and 9 summarize the final hyperparameter settings selected by the nested grid-search procedure for representative full-ring and compact calibration configurations. The RFBDT model combines a Random Forest baseline predictor with a LightGBM-based residual-correction model, whereas the BDT model corresponds to a standalone LightGBM regressor. Therefore, the two model families have partially different hyperparameter sets.

Table 8
Final hyperparameters selected by nested grid-search optimization for the RFBDT no-Bin configurations.

Hyperparameter	RFBDT no-Bin: Rings[1:100]	RFBDT no-Bin: PCA-5	Purpose
`rf_n_estimators`	1000	1000	Number of trees in the Random Forest baseline predictor. It increases ensemble stability and predictive capacity.
`rf_max_depth`	10	10	Maximum depth of each Random Forest tree. It limits model complexity and helps control overfitting.
`rf_min_samples_leaf`	24	24	Minimum number of samples required in a Random Forest leaf node. It regularizes the baseline predictor.
`rf_min_samples_split`	24	24	Minimum number of samples required to split an internal Random Forest node. It controls tree growth.
`rf_max_features`	1.0	1.0	Fraction of features considered when searching for the best Random Forest split. It controls feature subsampling.
`rf_max_samples`	0.7	0.7	Fraction of training samples used to fit each Random Forest tree under bootstrap sampling.
`rf_bootstrap`	True	True	Enables bootstrap sampling in the Random Forest baseline model.
`gb_num_leaves`	127	127	Maximum number of leaves in each LightGBM residual-correction tree. It controls model expressiveness.
`gb_min_child_samples`	24	24	Minimum number of samples required in a LightGBM leaf. It regularizes the residual model.
`gb_feature_fraction`	1.0	1.0	Fraction of features used by LightGBM at each boosting iteration. It controls feature subsampling.
`gb_bagging_fraction`	0.7	0.7	Fraction of samples used by LightGBM for bagging. It introduces stochasticity and helps reduce overfitting.
`gb_bagging_freq`	1	1	Frequency of LightGBM bagging. A value of 1 applies bagging at every boosting iteration.

(*continued on next page*)

Table 8 (*continued*).

Hyperparameter	RFBDT no-Bin: Rings[1:100]	RFBDT no-Bin: PCA-5	Purpose
gb_learning_rate	0.05	0.05	Shrinkage factor applied to each boosting step. It balances convergence speed and generalization.
gb_lambda_l2	0.0	0.0	L2 regularization applied to LightGBM leaf weights. It controls weight penalization in the residual model.
gb_n_estimators	1000	1000	Maximum number of boosting iterations in the LightGBM residual model.
val_fraction	0.1	0.1	Fraction of the training data internally reserved for validation during LightGBM residual-model training.
early_stopping_rounds	50	50	Stops boosting if validation performance does not improve after the specified number of rounds.
random_state	0	0	Seed used to ensure reproducibility of stochastic training procedures.
force_col_wise	True	True	Forces column-wise histogram construction in LightGBM, improving stability for tabular inputs.

Table 9
Final hyperparameters selected by nested grid-search optimization for the BDT no-Bin configurations.

Hyperparameter	BDT no-Bin: Rings[1:100]	BDT no-Bin: PCA-5	Purpose
boosting_type	gbdt	gbdt	Defines the boosting strategy. In this work, gbdt corresponds to Gradient Boosted Decision Trees implemented with LightGBM.
objective	regression	regression	Defines the learning objective used for continuous calibration-factor estimation.
n_estimators	1000	1000	Maximum number of boosted trees. It controls model capacity and convergence.
learning_rate	0.05	0.05	Shrinkage factor applied to each boosting step. It balances convergence speed and generalization.
max_depth	10	10	Maximum depth of each boosted tree. It limits model complexity and helps control overfitting.
num_leaves	127	255	Maximum number of leaves in each boosted tree. It controls the expressive capacity of the LightGBM model.
min_child_samples	24	32	Minimum number of samples required in a leaf. It regularizes leaf formation and reduces overly specific splits.
feature_fraction	1.0	1.0	Fraction of features used at each boosting iteration. It controls feature subsampling.
bagging_fraction	0.7	0.7	Fraction of samples used for bagging at each boosting iteration. It introduces stochasticity and helps reduce overfitting.
bagging_freq	1	1	Frequency of bagging. A value of 1 applies bagging at every boosting iteration.
lambda_l1	0.0	0.0	L1 regularization applied to leaf weights. It controls sparsity in the boosted model.
lambda_l2	0.0	0.0	L2 regularization applied to leaf weights. It controls weight penalization and improves stability.
min_gain_to_split	0.0	0.0	Minimum loss reduction required to perform a split. It controls conservative tree growth.
random_state	0	0	Seed used to ensure reproducibility of stochastic training procedures.
force_col_wise	True	True	Forces column-wise histogram construction in LightGBM, improving stability for tabular inputs.

Data availability

The data repository link is provided in the Appendix of the manuscript and is also included in the reference list.

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